

Readability Does Not Confer Control over Clinical Concepts

Hanqi Jiang^{1,2,*} Weihang You^{1,*} Qingchan Zhu¹ Junhao Chen¹ Yi Pan¹
Zhengliang Liu¹ Wei Liu³ Lei Xing^{2,†}

¹School of Computing, University of Georgia

²School of Medicine, Stanford University

³Department of Radiation Oncology, Mayo Clinic, Phoenix, AZ, USA

*Equal contribution.

†Corresponding author: lei@stanford.edu.

Preprint

Abstract

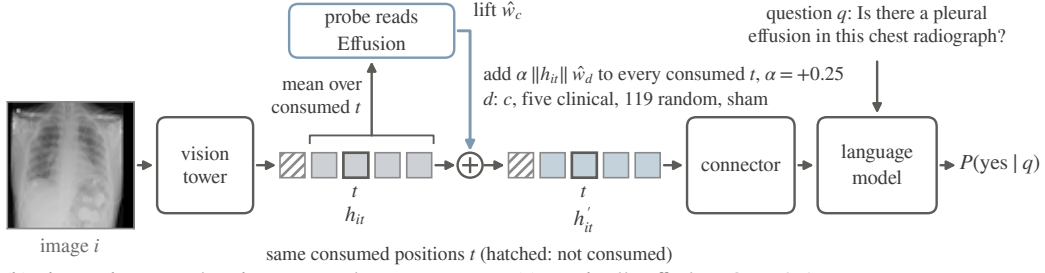
A linear probe reads a clinical finding from the visual representation that a vision-language model’s language model consumes, and activation steering suggests the converse: writing the probe direction back should move the model’s answer about the finding, making the direction a handle on the concept. Evaluations test whether models encode findings and answer questions about them, not whether such a handle exists. We test it with *ownership*: at a fixed dose on the same patients, does a concept’s own direction move its own answer more than the directions fitted for every competing finding? Across 25 checkpoints on NIH ChestX-ray14, CheXpert Plus and COCO, clinical concepts are readable in 59% of chest checkpoint–concept cells and answerable in 66%, but owned in 13%: most clinical writes do not move their own answer beyond random directions of the same dose, and the same directions move the other findings’ answers. Under the same procedure, 75% of natural-image object cells are owned, and the gap persists among cells the model answers well (30% against 87%). Non-clinical attributes of the radiographs share the deficit, and checkpoints that share a vision tower, and therefore write identical vectors, own different concepts: the deficit follows the chest-radiograph representation and the model that reads it. A direction fitted to the model’s own answers provides the handle that the label direction lacks. A probe’s readability does not confer control through its direction.

1 Introduction

Linear probes are the standard tool for testing what a vision-language model (VLM) represents. They read out a concept by fitting a classifier to activations at a chosen layer (Alain and Bengio, 2017; Belinkov, 2022). In medical imaging, probes often succeed at linearly decoding clinical findings such as pleural effusion or cardiomegaly from the visual representation of general-purpose and medical VLMs (Zhu et al., 2026; Pedersen et al., 2026). Activation steering promises the converse: writing the probe direction back into the representation to change the model’s behaviour (Turner et al., 2023; Panickssery et al., 2024). Together, these methods cast the probe direction as a handle on the concept: read it to detect, write it to control. We ask whether that handle exists.

A steering write that changes the answer does not settle the question. In Qwen2.5-VL-7B on NIH ChestX-ray14, a probe reads effusion from the visual tokens the language model consumes, and writing its direction back raises the model’s probability of answering yes to the Effusion question by 0.190, more than the 95th percentile of 119 random directions and a coordinate-permutation sham written at the same dose. By the usual steering criteria, the write works. Writing the Nodule direction into the same patients at the same dose raises the Effusion answer further, by 0.262 (Figure 1c), although Nodule is the competing direction least similar to Effusion (Appendix A.8). Random and sham directions carry no label, so they cannot expose a direction whose effect is shared with other

(a) read and write at the final visual block the language model consumes



(b) the grade: one write, three comparisons

(c) seed cell: Effusion, Qwen2.5-VL-7B, NIH

$W_{q,d}$: mean change in $P(\text{yes} | q)$ under the write of d

- own effect
 $W_{q,q} > 0$ ✓
- random and sham
 $W_{q,q} > \text{random p95 and } > \text{sham}$ ✓
- five clinical directions
 $W_{q,q} - W_{q,d} > 0$ for every $d \neq q$ ✗

owned: all three hold under simultaneous max- T bounds
seed cell: stronger competitor

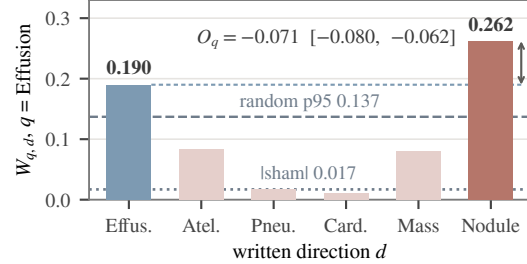


Figure 1: Reading and writing a concept at the consumed block. (a) The probe reads the concept from the consumed visual tokens, and its lifted normal is written back to them at dose $\alpha = +0.25$, a quarter of each token’s norm. (b) The write matrix $W_{q,d}$, the mean change in $P(\text{yes})$ of question q under a write along direction d , and the three comparisons that decide ownership. (c) The Effusion case (Qwen2.5-VL-7B, NIH ChestX-ray14): a stronger competitor.

findings (Pahde et al., 2025). Exposing it requires writing every competing clinical direction at the same locus and dose in the same patients and asking whether the concept’s own direction moves the concept’s own answer more than any other. We call the outcome *ownership*. When a concept is readable but not owned, its probe direction reads the concept without writing it: reading is not writing.

We turn this comparison into an evaluation (Section 3, Figure 1). At the *consumed block*, the final visual block whose output the language model reads, each concept receives three grades. It is *readable* if its probe beats random-label probes of equal capacity, *answerable* if the model’s yes/no answer without any write beats chance, and *owned* if a fixed-dose write along its direction clears the random and sham references and moves its own answer more than every competing clinical direction under simultaneous bounds. We apply the evaluation to 25 checkpoints from twelve families (3B–72B; general-purpose and medical) on two chest-radiograph datasets with report-derived labels, NIH ChestX-ray14 and CheXpert Plus, and on COCO objects as a natural-image control (Section 4).

On chest radiographs most readable and answerable concepts are not owned, while most COCO objects are, and the gap persists among concepts the model answers well (Section 5). We call this gap the *attribution deficit*. Since the same write succeeds on natural images, the remaining results locate the deficit. Checkpoints that share a vision tower write identical vectors into identical representations yet own different concepts, so the model that reads the representation decides ownership. Non-clinical attributes of the same radiographs (projection, sex, and age) are owned no more often than findings, so the deficit does not depend on the concept being clinical. The reader responds to a different direction: a direction fitted to the model’s own answers is owned in 19 of 36 chest checkpoint–concept cells where the label direction is owned in 9, and on chest radiographs it is nearly orthogonal to the probe direction.

Contributions. (i) We introduce ownership, which separates reading a concept from writing it: a probe direction is scored against directions fitted the same way for the competing concepts, written at the same locus and dose in the same patients (Section 3). (ii) We identify the attribution deficit:

across 450 checkpoint–concept cells, clinical concepts on chest radiographs are read and answered far more often than they are owned, and the gap to natural-image objects persists at matched answer ability (Sections 5.1–5.3). (iii) We trace the deficit to how the model reads the chest-radiograph representation rather than to the concept’s clinical status, and show that a direction fitted to the model’s own answers provides the handle that the label direction lacks (Sections 5.4–5.6). We release the evaluation, scorecards, and write matrices; a new model needs one adapter naming its consumed block.

2 Background and Related Work

Probing and its controls. Linear probes track information across network depth (Alain and Bengio, 2017), and a probe score reflects both the representation and the learner fitted to it, which motivates matched-capacity probes and control tasks (Hewitt and Liang, 2019). The probing literature distinguishes information a readout can recover from information the model uses (Belinkov, 2022; Ravichander et al., 2021); causal erasure tests use by deleting a concept (Ravfogel et al., 2020; Elazar et al., 2021), in closed form for every linear reader at once (Belrose et al., 2023). Our readability grade adopts the control-task idea, and the write test asks a question complementary to erasure: whether the concept’s own direction is the one that moves its answer. In medical multimodal models, depth-wise probing maps classification information across towers, connectors, and language layers (Zhu et al., 2026), and layer-wise probes of MedCLIP expose acquisition and device shortcuts (Pedersen et al., 2026); we intervene on the decoded clinical direction at the locus where it is read.

Concept directions and steering. A concept activation vector is the normal of a classifier separating concept examples from a negative set, and it grades a class by the directional derivative of the logit along that normal (Kim et al., 2018); we write the direction forward instead. Such a normal shifts with the layer it is fitted at, entangles with correlated concepts, depends on its negative set, and carries where in the image the concept was read (Nicolson et al., 2025). Representation engineering and activation steering write a concept back by changing intermediate representations at inference time (Zou et al., 2023; Turner et al., 2023; Panickssery et al., 2024), and a probe normal is an appealing intervention because it supplies a label and a signed vector from the same representation. A separability-trained vector can, however, absorb distractor signals and diverge from its concept (Pahde et al., 2025): a classifier’s weights are a filter rather than the pattern they detect (Haufe et al., 2014), mass-mean directions steer truth judgements better than logistic normals (Marks and Tegmark, 2024), and the features that identify a concept are largely not those that control the output (Arad et al., 2025). Benchmarks accordingly score concept detection and steering as separate axes (Wu et al., 2025), and alignment search certifies a subspace as causal when it implements an interpretable variable (Geiger et al., 2024). These works judge a direction mainly against its own concept. Ownership fixes the estimator and varies the label instead: it asks whether a direction fitted the same way for another concept moves the same answer more. Section 5.7 runs alternative constructions, a second locus, and token-concentrated writes through this test.

Writing into a vision-language model. Image features are linearly decodable in a vision-language model’s hidden layers, and targeted edits to them change downstream answers (Rajaram et al., 2025). Steering one sparse-autoencoder feature of the vision tower and asking the language model what it sees recovers what that feature encodes (Ferrando et al., 2026); we start from the meaning, fit a direction to a clinical label, and ask whether the write keeps that label’s selectivity against the other findings’ directions. Causal probing of multimodal models intervenes on the language model’s residual stream with difference-in-means vectors and scores each concept against its own effect (Deng et al., 2026); we write the representation the language model consumes and score each concept against its competitors.

3 Reading, Answering, and Writing at the Consumed Block

We ask three questions about a concept c in a checkpoint M on a dataset with binary labels: whether c is readable from the visual representation the language model consumes, whether M answers the question about c , and whether writing c 's own direction moves c 's answer more than any alternative. A registered protocol fixes an estimator and a decision rule for each question (Appendix A.1).

3.1 Consumed block and matched read–write positions

Every VLM in the grid passes an image through a vision tower and hands a specific tensor to a connector or merger that produces the language model's inputs. The *consumed block* is the last visual block whose output that consumer reads, restricted to the token positions the consumer actually uses (CLS or padding rows are excluded where the consumer drops them); the *connector locus* is the consumer's output. Appendix F.1 identifies the block for every family. For image i and consumed token t , let $\mathbf{h}_{it} \in \mathbb{R}^D$ be the output of the consumed block, where D is the tower's hidden width; mean pooling over the consumed tokens gives the read vector $\mathbf{h}_i$. A write adds the unit direction $\hat{\mathbf{w}}_c$ of concept c (Section 3.2) to the same tokens,

$$\mathbf{h}'_{it} = \mathbf{h}_{it} + \alpha \|\mathbf{h}_{it}\|_2 \hat{\mathbf{w}}_c, \quad (1)$$

so that the dose α is a fraction of each token's own norm, comparable across families whose activation scales differ by orders of magnitude, and its sign selects the concept's positive or negative pole. Reading and writing therefore use the same representation and the same positions. The connector and the language model, everything downstream of the consumed block, form the *reader* of that representation.

3.2 Directions, controls, and the reference family

Each concept is read by a linear probe on a shared 512-dimensional view of the representation. A fixed Gaussian matrix $R \in \mathbb{R}^{D \times 512}$ maps the read vector to features $\mathbf{x}_i = R^\top \mathbf{h}_i$, which are standardised with training-row means and scales s ; an ℓ_2 -regularised logistic regression with coefficient vector β_c then scores concept c (inverse regularisation strength 1, 2,000 iterations, three fit seeds). The probe's logit is thus an affine function of $\mathbf{h}_i$ whose gradient is $R \text{diag}(s)^{-1} \beta_c$, and the write direction is that gradient at unit length,

$$\hat{\mathbf{w}}_c = \frac{\mathbf{g}_c}{\|\mathbf{g}_c\|_2}, \quad \mathbf{g}_c = R \text{diag}(s)^{-1} \beta_c. \quad (2)$$

It is the direction in the consumed representation along which the probe's evidence for c grows fastest (Appendix F.3 gives the numerical details). We call it the concept's *label direction*, since it is fitted to the concept's labels.

Two families of controls bracket the label direction. For reading, 20 control probes are fitted to random labels drawn within recurring image types (view, sex, and age decade on chest sets; aspect and area buckets on COCO), so that they have the capacity and the nuisance structure of the concept probe. With y the concept labels and r the probe's scores on the calibration rows, and y_k^{ctrl} and r_k^{ctrl} the random labels and scores of control probe k , the probe's selectivity is

$$S = \text{AUROC}(y, r) - \frac{1}{20} \sum_{k=1}^{20} \text{AUROC}(y_k^{\text{ctrl}}, r_k^{\text{ctrl}}), \quad (3)$$

which is positive when the probe reads its concept better than a probe of equal capacity reads an arbitrary label with the same nuisance structure. For writing, the *reference family* contains 119 random unit directions, a coordinate-permutation sham of $\hat{\mathbf{w}}_c$ (same coordinates, scrambled arrangement), and the five label directions of the other clinical concepts, all written at the same dose to the same images.

3.3 Ownership

Each dataset has six concepts, and we index both the yes/no question about a concept and the direction fitted for it by that concept. For question q and direction d , $W_{q,d}$ is the mean over test images of the change in $P(\text{yes})$ for question q when $\hat{\mathbf{w}}_d$ is written, each image scored with and without the write. Here $P(\text{yes}) = \sigma(\ell_{\text{yes}} - \ell_{\text{no}})$, where σ is the logistic function and ℓ_{yes} and ℓ_{no} are the log-probabilities of the two answers at the first answer position, each the maximum over case and leading-space variants that tokenise to one token. Row q of the 6×6 write matrix W records how far each direction moves question q , and its diagonal entry $W_{q,q}$ is the concept’s own write (Figure 5). Ownership is the diagonal margin

$$O_q = W_{q,q} - \max_{d \neq q} W_{q,d}, \quad (4)$$

the advantage of the concept’s own direction over its strongest clinical competitor on the concept’s own question. $W_{q,q}$ measures how far the concept’s own direction moves its answer; O_q measures whether it moves that answer further than every competitor built the same way. A large $W_{q,q}$ shows that $d = q$ can write the answer; only $O_q > 0$ shows that the label names the effective direction. A concept can therefore fail ownership in two ways: its own write stays within the random and sham references, or the write clears them without holding its advantage over every competitor. Section 5.1 measures both. The Effusion case of the introduction illustrates the latter: the Effusion direction raises the Effusion answer by 0.190, above the random 95th percentile and the sham, and the Nodule direction raises the same answer by 0.262, so $O_{\text{Effusion}} = -0.071$. Uncertainty over patients comes from 2,000 bootstrap draws of test rows, with the maximum in Equation 4 recomputed inside every draw. The 30 contrasts $W_{q,q} - W_{q,d}$ (six questions, five competitors each) receive simultaneous 95% bounds from a centred studentised max- T bootstrap, which hold jointly across all 30 (Appendix E).

3.4 Graded outcomes

The readable and answerable grades are computed on the 400 calibration rows, which carry no write. The write matrix and every ownership quantity are computed on the 600 test rows, which share no patient with the calibration rows.

- **Readable:** the one-sided 95% lower bound of S on the calibration rows is above zero, with at least ten known positives and ten known negatives among those 400 rows.
- **Answerable:** the *clean answer*, $P(\text{yes})$ with no write, separates the concept’s labels: the one-sided 95% lower bound of its AUROC on the same calibration rows is above 0.5, under the same ten-and-ten support rule.
- **Owned:** on the test rows, the write meets the *steering reference* ($W_{q,q} > 0$, above the 95th percentile of the 119 random writes, and above the sham’s absolute effect) and every simultaneous lower bound of $W_{q,q} - W_{q,d}$ is positive. A cell with any simultaneous upper bound below zero has a *stronger competitor*; the rest are *unresolved*.

A checkpoint–concept pair on one dataset is a *cell*, and a checkpoint–dataset pair is a *block*. A cell graded owned is said to be owned, and the checkpoint is said to own the concept; an alternative direction is owned in a cell when its write meets the same rule for that concept. A block contributes to the answerable and owned counts only when its write matrix and its calibration pass both completed. The grades are defined for one direction construction, one primary dose with a five-point sweep, six templates, pooled consumed-token features, and two loci, the consumed block and the connector output. Before a block is scored, seven acceptance checks on 16 held-out preflight rows confirm, at both loci, that scoring is deterministic, that $\alpha = 0$ changes nothing, that the write reaches the answer and changes nothing outside the consumed tokens, and that every template maps its answers correctly without an image (Appendix A.3).

4 Models, Datasets, and Scale

Checkpoints. We evaluate 25 instruction-tuned checkpoints from twelve families, 3B to 72B (Table 2). Six families are general-purpose: Qwen2.5-VL (Bai et al., 2025b), Qwen3-VL (Bai et al., 2025a), InternVL3.5 (Wang et al., 2025), Gemma 3 (Gemma Team, 2025), LLaVA-1.5 (Liu et al., 2024), and Llama 3.2 Vision (Meta, 2024). Six are medical fine-tunes: MedGemma (Sellersgren et al., 2025), Lingshu (LASA Team et al., 2025), LLaVA-Med 1.5 (Li et al., 2023), CheXagent (Chen et al., 2024b), HuatuoGPT-Vision (Chen et al., 2024a), and LLaVA-Rad (Zambrano Chaves et al., 2025). They span six vision-tower designs and three consumer designs: token insertion after a merger, token insertion after a projector, and cross-attention. Three towers are trained on chest radiographs, those of CheXagent-2-3B, CheXagent-8B, and LLaVA-Rad-7B (Appendix A.2). Every checkpoint runs in bf16 at a pinned revision with its native processor.

Datasets and cohorts. NIH ChestX-ray14 (Wang et al., 2017) and CheXpert Plus (Chambon et al., 2024) provide frontal chest radiographs with report-derived labels, the CheXpert labels coming from the CheXpert labeler (Irvin et al., 2019). We evaluate six findings per dataset (NIH: Effusion, Atelectasis, Pneumothorax, Cardiomegaly, Mass, Nodule; CheXpert: Effusion, Atelectasis, Pneumothorax, Cardiomegaly, Consolidation, Edema). COCO (Lin et al., 2014) provides the natural-image control, with binary presence labels for six object categories (person, dog, car, chair, bottle, bicycle). Seeded hashes of patient or image identity select one fixed cohort per dataset, shared by every checkpoint: 20,000 training rows (18,212 on NIH) fit the probes and the answer direction, 400 calibration rows carry the readable and answerable grades, 600 test rows carry every write, and 16 preflight rows carry the acceptance checks. No patient and no image appears in two roles of a dataset (Appendix F.5).

Questions and scale. Each concept is asked as a yes/no question in six templates that cross two wordings (*is there* versus *does the image show*) with three answer interfaces (yes/no, A-present, B-present); the *is/yes-no* template gives the primary grade (Appendix D). The primary dose is $\alpha = +0.25$, and a sweep adds $\alpha \in \{-0.5, -0.25, -0.1, +0.1, +0.5\}$. Each block scores the write matrix, the sham, and the 119 random directions on the 600 test rows. The campaign comprises 75 blocks, 450 scored concept cells, about 30 million scored answers, and roughly 3,000 A100-hours; 74 of the 75 blocks carry every planned analysis (Appendix A.4).

5 Results

5.1 Reading is not writing

On both chest datasets, reading and answering far outpace writing (Table 1, Figure 2a–b). Clinical concepts are readable in 178 of 300 cells (59%; median selectivity 0.14), and 199 (66%) are answerable. Yet only 38 cells (13%) are owned, and reading and answering together barely change this: 28 of the 141 readable and answerable cells are owned (20%), against 10 of the remaining 159 (6%).

Ownership on chest radiographs breaks at the write itself. 244 of 300 label directions fail the random and sham reference for their own question, and these cells carry 187 of the 196 stronger-competitor verdicts (95%; Table 8). Once an own write clears the reference, 38 of 56 cells are owned; on COCO, 117 of 150 writes clear it and 113 are owned. On chest radiographs, most label directions therefore scarcely move the answers they name, while the directions of other findings often move those answers more.

The Effusion case reproduces on 600 new patients (Table 24). The Effusion write raises its answer by 0.190, second among the 120 compared effects, the Nodule write raises it by 0.262, and $O_{\text{Effusion}} = -0.071 [-0.080, -0.062]$, matching the $-0.057 [-0.069, -0.045]$ of the two-model pilot study that motivated the evaluation (Appendix B). Across the grid, Effusion is owned in 7 of 50 chest blocks, and in 6 of these $O_{\text{Effusion}} \leq 0.03$.

Table 1: **Reading, answering, and writing per checkpoint.** Per dataset: mean probe selectivity S (READ), mean clean-answer AUROC (ANS), and mean ownership O_q (OWN), each superscripted with the number of the six concepts that receive the grade; then the owned share of clinical and COCO cells and their ratio, by which rows are sorted. Shading darkens with magnitude (blue for negative OWN); “–” marks an undefined ratio.

Checkpoint	NIH ChestX-ray14			CheXpert Plus			COCO (control)			Owned share		
	READ	ANS	OWN	READ	ANS	OWN	READ	ANS	OWN	clin.	COCO	ratio
Lingshu 32B	0.08 ³	0.78 ⁶	-0.02 ²	0.20 ⁴	0.87 ⁵	+0.04 ⁵	0.03 ¹	0.95 ⁵	+0.28 ⁶	58%	100%	58%
CheXagent-2-3B	0.13 ⁵	0.86 ⁶	+0.00 ²	0.21 ⁵	0.95 ⁵	+0.01 ⁴	0.13 ⁴	0.81 ⁵	+0.03 ²	50%	33%	150%
Qwen2.5-VL-72B	0.07 ²	0.62 ⁴	-0.03 ³	0.20 ⁴	0.71 ⁴	-0.09 ¹	0.01 ¹	0.99 ⁵	+0.66 ⁶	33%	100%	33%
Lingshu 7B	0.08 ³	0.76 ⁶	+0.02 ¹	0.21 ⁴	0.86 ⁵	+0.04 ³	0.02 ¹	0.97 ⁵	+0.68 ⁶	33%	100%	33%
Qwen3-VL-4B	0.08 ³	0.71 ⁶	-0.11 ⁰	0.20 ⁴	0.74 ⁴	-0.01 ²	0.04 ⁴	0.99 ⁵	+0.34 ⁶	17%	100%	17%
Qwen3-VL-32B	0.10 ²	0.70 ⁶	-0.02 ¹	0.17 ⁵	0.82 ⁵	-0.02 ¹	0.05 ⁴	0.99 ⁵	+0.15 ⁶	17%	100%	17%
InternVL3.5-38B	0.10 ⁵	0.78 ⁶	-0.00 ⁰	0.17 ⁴	0.86 ⁵	-0.00 ²	0.08 ⁴	0.99 ⁵	+0.01 ⁶	17%	100%	17%
Gemma 3 12B	0.07 ²	0.62 ⁵	-0.37 ¹	0.18 ⁴	0.52 ⁰	-0.38 ¹	0.07 ⁵	0.95 ⁵	+0.44 ⁶	17%	100%	17%
MedGemma 4B	0.12 ³	0.80 ⁶	-0.13 ⁰	0.21 ⁴	0.87 ⁵	-0.16 ²	0.06 ⁴	0.94 ⁵	+0.23 ⁶	17%	100%	17%
InternVL3.5-14B	0.07 ⁴	0.73 ⁶	-0.00 ¹	0.16 ⁴	0.87 ⁵	+0.00 ¹	0.12 ⁴	0.98 ⁵	+0.02 ⁵	17%	83%	20%
Qwen2.5-VL-32B	0.09 ³	0.63 ⁴	-0.06 ¹	0.16 ⁴	0.70 ⁴	-0.11 ⁰	0.02 ¹	0.99 ⁵	+0.38 ⁶	8%	100%	8%
InternVL3.5-8B	0.06 ²	0.73 ⁶	-0.01 ⁰	0.19 ⁴	0.86 ⁵	-0.00 ¹	0.12 ⁵	0.98 ⁵	+0.03 ⁶	8%	100%	8%
HuatuoGPT-Vision-7B	0.08 ³	0.71 ⁶	-0.03 ¹	0.19 ⁴	0.78 ⁴	-0.01 ⁰	0.07 ⁴	0.99 ⁵	+0.01 ⁴	8%	67%	12%
LLaVA-Med 1.5 7B	0.08 ³	0.53 ⁰	-0.01 ⁰	0.19 ⁴	0.59 ²	-0.01 ¹	0.07 ⁴	0.76 ⁵	+0.00 ¹	8%	17%	50%
CheXagent-8B	0.16 ⁵	0.73 ⁶	-0.00 ⁰	0.23 ⁵	0.88 ⁵	-0.02 ¹	0.05 ³	0.52 ¹	-0.03 ⁰	8%	0%	–
Qwen2.5-VL-3B	0.07 ²	0.58 ²	-0.08 ⁰	0.17 ⁴	0.57 ²	-0.07 ⁰	0.05 ⁴	0.97 ⁵	+0.52 ⁶	0%	100%	0%
Qwen2.5-VL-7B	0.08 ³	0.62 ³	-0.14 ⁰	0.20 ⁴	0.65 ²	-0.08 ⁰	0.02 ¹	0.98 ⁵	+0.67 ⁶	0%	100%	0%
Qwen3-VL-8B	0.10 ⁴	0.69 ⁶	-0.01 ⁰	0.17 ⁴	0.75 ⁴	-0.04 ⁰	0.04 ³	0.99 ⁵	+0.58 ⁶	0%	100%	0%
Gemma 3 27B	0.07 ²	0.60 ³	-0.03 ⁰	0.18 ⁴	0.55 ²	-0.01 ⁰	0.07 ⁵	0.96 ⁵	+0.31 ⁶	0%	100%	0%
Gemma 3 4B	0.07 ²	0.56 ²	-0.11 ⁰	0.18 ⁴	0.53 ¹	-0.27 ⁰	0.07 ⁵	0.94 ⁵	+0.24 ⁵	0%	83%	0%
LLaVA-1.5-7B	0.08 ²	0.51 ⁰	-0.04 ⁰	0.17 ⁴	0.52 ⁰	-0.03 ⁰	0.28 ⁵	0.97 ⁵	+0.03 ⁵	0%	83%	0%
MedGemma 27B	0.12 ³	0.80 ⁶	-0.14 ⁰	0.21 ⁴	0.87 ⁵	-0.06 ⁰	0.06 ⁴	0.97 ⁵	+0.12 ⁴	0%	67%	0%
LLaVA-1.5-13B	0.08 ²	0.53 ⁰	-0.02 ⁰	0.17 ⁴	0.51 ¹	-0.01 ⁰	0.28 ⁵	0.96 ⁵	+0.02 ²	0%	33%	0%
Llama 3.2 Vision 11B	0.08 ²	0.65 ⁴	-0.03 ⁰	0.16 ⁴	0.73 ⁴	-0.04 ⁰	0.12 ⁵	0.95 ⁵	-0.01 ¹	0%	17%	0%
LLaVA-Rad-7B	0.15 ⁴	0.71 ⁵	-0.01 ⁰	0.27 ⁵	0.87 ⁵	-0.02 ⁰	0.12 ⁴	0.50 ⁰	-0.03 ⁰	0%	0%	–
All cells	0.09	0.68	-0.06	0.19	0.74	-0.05	0.08	0.92	+0.23	13%	75%	17%
count / cells	74/150	110/150	13/150	104/150	89/150	25/150	90/150	116/150	113/150			

5.2 Where the chest write goes

Clinical writes are not selective (Figure 3). The written concept’s own question receives 24% of the positive answer change on NIH and 22% on CheXpert, against 79% on COCO. The write matrices show the same structure (Figure 5). The own write is small, with median $W_{q,q}$ of 0.010 on NIH and 0.020 on CheXpert against 0.127 on COCO, and the competing directions share what effect there is: the Atelectasis, Effusion, and Consolidation directions raise each other’s answers by similar amounts, and on the chest sets a row of W is flat or peaks off the diagonal, whereas on COCO the diagonal dominates. At every Qwen2.5-VL size a competing direction raises the Effusion answer more than the Effusion direction does (Cardiomegaly at 3B, Nodule at 7B and 32B, Atelectasis at 72B). Single images show the pattern (Figure 4): on a radiograph, the Effusion write raises the Effusion answer from 0.02 to 0.56 and the Nodule write raises it further, to 0.75, while on a natural image the dog write moves the dog answer from 0.00 to 1.00 and nothing else. The chest cells that are owned are also fragile: an effect floor of 0.05 on both $W_{q,q}$ and O_q keeps 18 of the 38 owned chest cells against 86 of the 113 on COCO, and ownership survives both probe refits in 4 of 13 owned NIH cells against 99 of 113 on COCO (Table 13).

5.3 The write mechanism is intact on natural objects

On COCO, the same procedure yields the opposite result (Table 1, right block). Object ownership holds in 113 of 150 cells (75%), 117 cells meet the steering reference, only 17 have a stronger com-

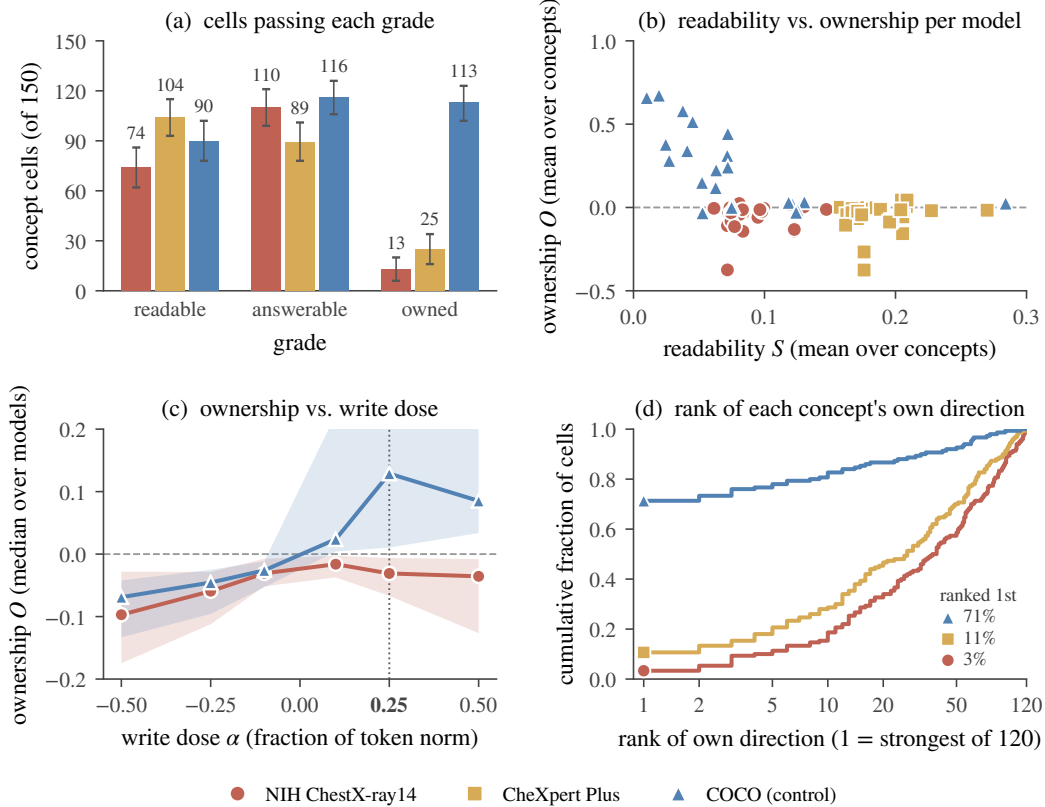


Figure 2: **Reading, answering, and writing across the grid.** (a) Readable, answerable, and owned cells of 150 per dataset, with descriptive 95% bootstrap intervals over cells. (b) Mean readability against mean ownership per checkpoint. (c) Median ownership and interquartile band over 25 blocks against the dose; the dotted line marks the graded dose $\alpha = +0.25$, and the COCO band is clipped at 0.2. (d) Cumulative share of cells by the rank of the concept write among the 120 compared effects.

petitor, and 19 of the 25 checkpoints own at least four of the six objects. Ownership also tracks the dose: on COCO, median ownership rises from -0.03 at $\alpha = -0.1$ to $+0.13$ at $\alpha = +0.25$ and stays positive at $\alpha = +0.5$ ($+0.09$), while on NIH it stays between -0.097 and -0.016 at every dose (Figure 2c). The write, direction construction, reference family, and decision rules thus work as intended.

The contrast survives two difficulty controls. Six small or fine-grained COCO categories (hand-bag, book, backpack, traffic light, knife, and cell phone) remain owned in 30 of 36 cells, against 34 of 36 for the easy six in the same 6 checkpoints (sign test $p = 0.25$), at a median clean-answer AUROC of 0.963 against 0.980 (Table 23). On the chest sets, ownership also stays rare where the model already answers the question well: at a clean-answer AUROC of at least 0.90, 11 of 37 chest cells are owned (30%) against 141 of 162 natural-image cells (87%), a difference of -0.573 [$-0.763, -0.388$] (Appendix A.10 states the category rule, the bins, and the bootstrap).

5.4 The reader decides ownership

Checkpoints that share a vision tower differ in what they own. Gemma 3 shares one SigLIP tower across its 4B, 12B, and 27B checkpoints, MedGemma one MedSigLIP tower across 4B and 27B, and LLaVA-1.5 one CLIP tower across 7B and 13B. Within each family, pooled features, fitted probes, and lifted write vectors are identical across sizes (Table 14), so readability is identical by construction and only the connector and language model that read the written representation differ. Steering a feature of the vision tower and letting the language model say what it sees treats that language model as a stable read-out of the tower (Ferrando et al., 2026); here the read-out is what varies. In 11 of

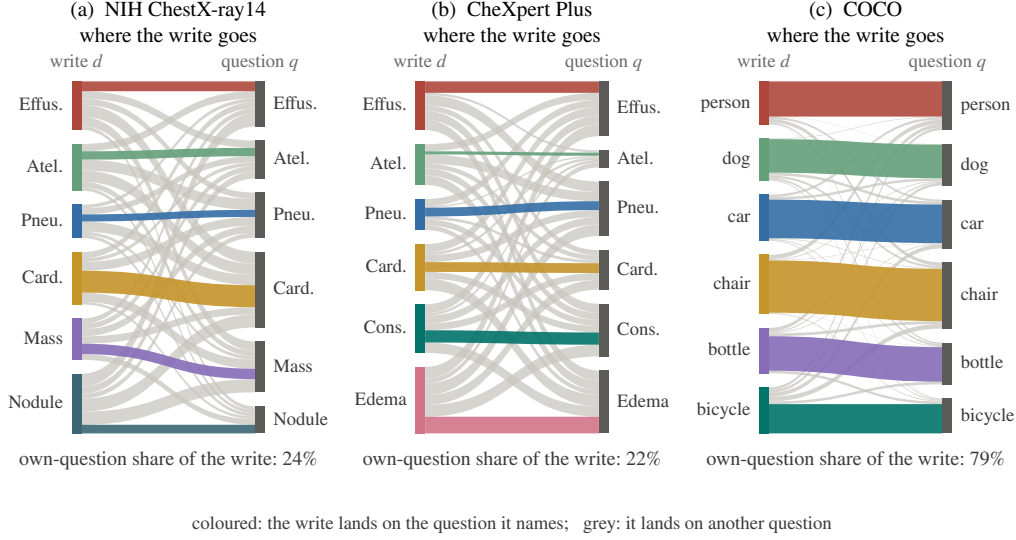


Figure 3: **Where the write goes.** Bands join each written direction d to the questions q it moves, with width proportional to $F_{q,d}$, the mean over checkpoints of $\max(W_{q,d}, 0)$. The percentage is the own-question share $\sum_d F_{d,d} / \sum_{q,d} F_{q,d}$.

the 12 shared-tower pair–dataset combinations, the paired patient-bootstrap interval for the between-reader difference ΔO_q excludes zero for five or six of the six concepts (Figure 6). On NIH Effusion, the identical write on the same 600 patients yields an O_q difference of -0.83 $[-0.85, -0.81]$ between Gemma 3 4B and 12B. In 12B, competing directions win the chest answers outright ($O_{\text{Effusion}} = -0.83$ on both chest sets), and LLaVA-1.5 7B and 13B own no chest concept yet own 5 and 2 COCO objects with the same write (Appendix A.9).

A better chest representation does not supply the missing reader. The three checkpoints whose towers are trained on chest radiographs read and answer the findings best in the grid (median probe AUROC 0.855 on NIH against 0.774 for the other NIH cells; median clean-answer AUROC 0.837 against 0.797 and 0.652 for medically tuned and general checkpoints), yet they own 7 of their 36 chest cells, against 15 of 72 for medically tuned checkpoints on general towers and 16 of 192 for general checkpoints. Scale raises chest ownership within some families, from none at Qwen2.5-VL 3B and 7B to three NIH concepts at 72B and from three to five CheXpert concepts across Lingshu 7B and 32B, but answering alone does not produce it: InternVL3.5 answers every NIH question at every size while its chest writes stay within 0.042 of zero (Figure 7). Ownership is therefore a property of the checkpoint that reads the representation, and a probe fitted to the tower alone cannot certify it.

5.5 Non-clinical attributes of the radiograph share the deficit

A non-clinical property of the same film is owned as rarely as a finding. We combine projection, sex, age, and the findings into one nine-direction family. The attribute probes read their labels at least as well as the clinical probes read the findings (median probe AUROC 0.896–0.999 against 0.808), yet attribute cells are owned in 12 of 114 cases and finding cells in 23 of 228, the same rate. At a clean-answer AUROC of at least 0.90, attributes are owned in 0 of 13 cells and findings in 6 of 24 (Table 17). The deficit therefore follows the chest-radiograph representation and its reader rather than the concept’s clinical status.

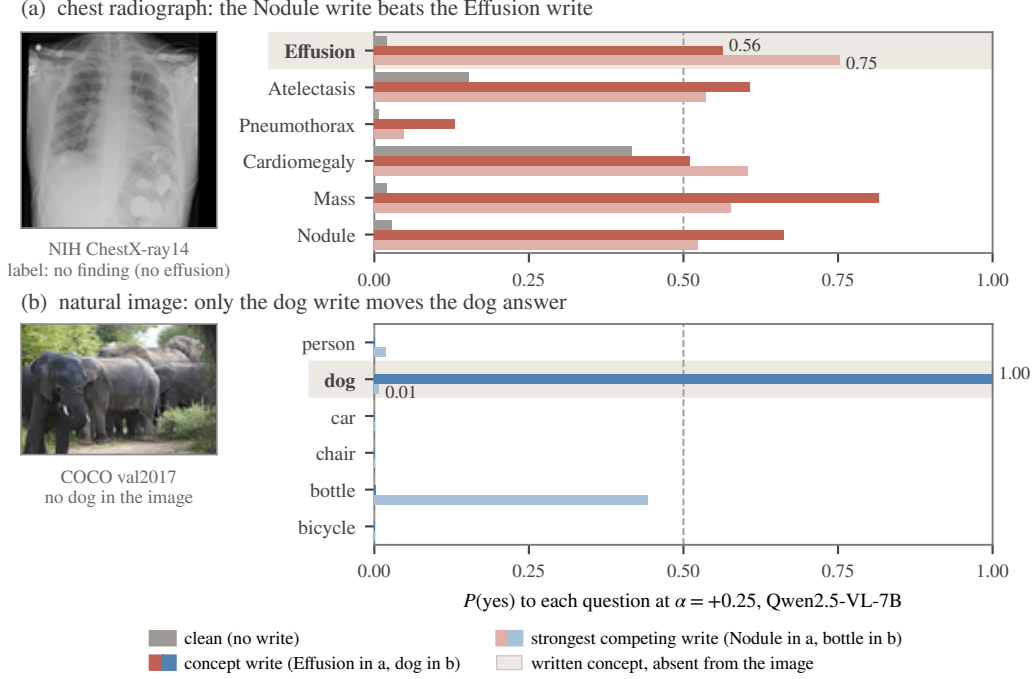


Figure 4: **Two images, six questions, three writes.** $P(\text{yes})$ per question for an NIH radiograph (top) and a COCO image (bottom), clean and under the concept write and the strongest competing write ($\alpha = +0.25$, Qwen2.5-VL-7B). The shaded row is the written concept, absent from the image; the dashed line marks $P(\text{yes}) = 0.5$.

5.6 The direction the reader responds to

For each question we fit the model’s own clean answer margin $\ell_{\text{yes}} - \ell_{\text{no}}$ on training rows from the same projected, train-scaled features by ridge regression, lift the fitted coefficients with Equation 2, and write the result on the test rows exactly as we write a label direction (Table 16). This *answer direction* a_q is a handle where the label direction is not. Across the 36 chest cells of 6 blocks, a_q is owned in 19 and the label direction in 9, and a_q remains owned in 76 of 95 chest (cell, template) pairs under the five held-out templates. The features that identify a concept differ from the features that control the output (Arad et al., 2025); here that separation holds against matched clinical competitors.

The answer direction reads the model, not the label. Over the 36 chest cells, its median AUROC against the report label is 0.66, against 0.80 for the probe, while against the model’s own clean answer it is 0.87, against 0.68 (Table 19). The two directions are far apart on chest radiographs, with a median raw model-space cosine of 0.18, and substantially aligned on natural images (0.59). The separation persists at four endpoints that fit neither direction (Table 22): the negated question, a forced choice against the strongest competitor in each of its two presentation orders, and the finding word’s log-probability in a report continuation. Across the 72 endpoint cells, the answer direction is owned in 30 and the label direction in 10. On chest radiographs, the direction the model answers along is not the direction the probe fits.

5.7 Controls and alternatives

The reference family behaves as designed: owned concept writes rank 1–6 among the 120 compared effects, and the concept write typically ranks in the top decile on the chest sets (Figure 2d; Table 9). The deficit survives every alternative we tested; Table 10 gives the rows, reference, rule, and sham of each of the 20 graded analyses, and Appendices A.8 and A.9 give the details.

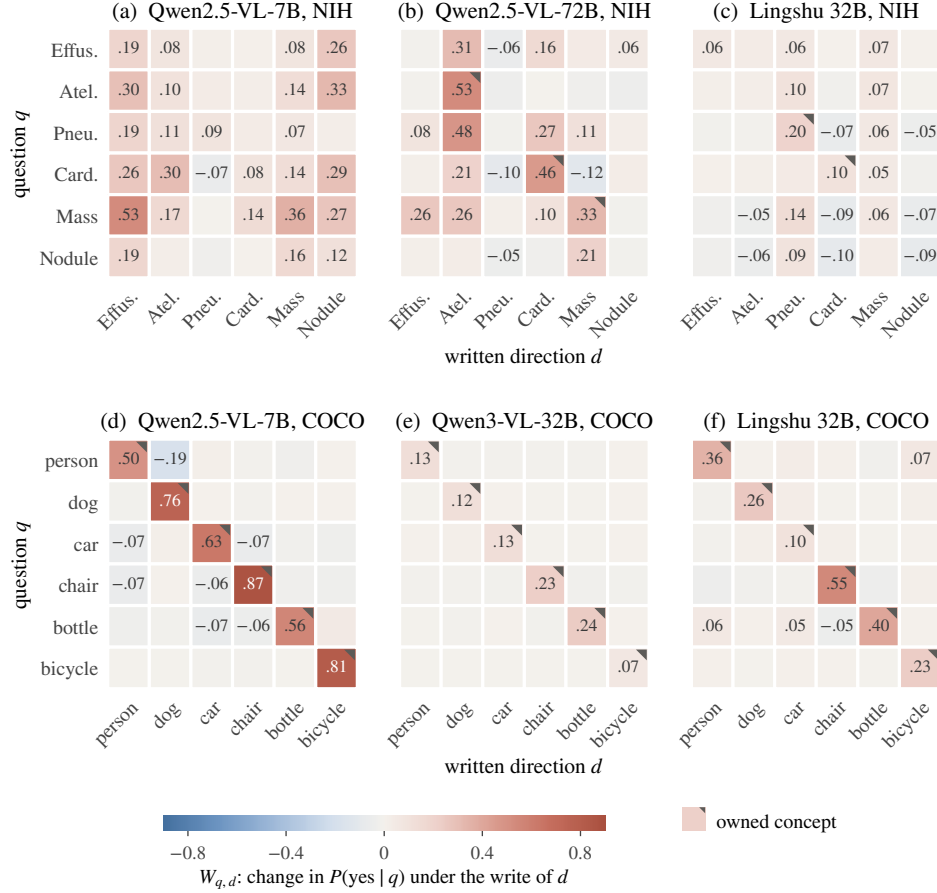


Figure 5: **Write matrices.** $W_{q,d}$, the mean change in $P(\text{yes})$ of question q (rows) under the write of direction d (columns), for three checkpoints on NIH (top) and three on COCO (bottom). A corner mark flags owned concepts.

Geometry, scale, and numerics. The strongest competitor matches the most similar direction or the most co-occurring label only at chance rate (Table 11). The logit-scale grade agrees with the probability-scale grade in 440 of 450 cells; restricting each cell to unsaturated rows preserves the owned grade in 36 of 38 owned chest cells; refitting the probes on resampled training patients changes O_q by a median of 0.02; and rescoring in fp32 at batch size one alters 1 of 108 grades (Table 12).

Labels. Re-grading the fitted directions on 200 CheXpert films labelled by radiologist consensus (Irvin et al., 2019) reproduces the owned grade in 112 of 114 cells, and directions refitted on those labels are owned in 5 of 36 cells, against 12 for the report-label directions (Table 20).

Directions. Four alternative constructions, the difference of class means, the Haufe pattern (the training covariance applied to the probe weights; Haufe et al., 2014), the normal orthogonalised against the other five normals, and the normal refitted on features residualised against the other five labels (Nicolson et al., 2025), are each owned in 6–16% of a chest dataset’s cells against 44–85% of COCO cells. Extending the competitor family to every additional dataset label with enough support retains 8 of the 11 owned cells, and writes concentrated on the tokens the probe reads are owned in 0 (softmax-weighted) and 5 (top quarter) of 36 chest cells, against 8 for the uniform write (Table 15). Concentrating the write where the probe reads does not create a clinical handle.

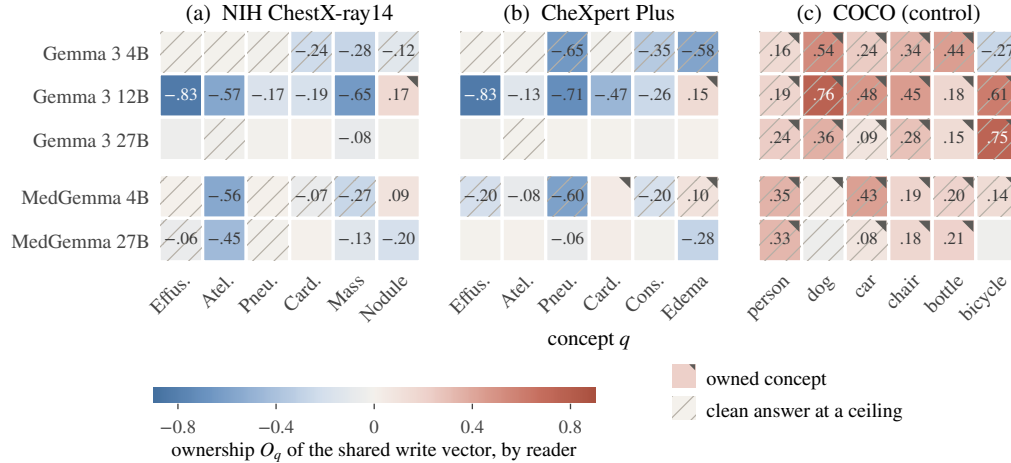


Figure 6: **Same write, different reader.** Ownership O_q per concept on NIH, CheXpert, and COCO for Gemma 3 4B/12B/27B and MedGemma 4B/27B, whose probes and write vectors are identical across sizes within each family. Corner marks flag owned cells and hatching marks ceiling cells.

6 Discussion

Readability belongs to the probe, ownership to the reader. Readability describes a classifier fitted to the representation; ownership describes how the model reading that representation responds to a write. Checkpoints sharing a vision tower have identical probes and write vectors, yet own different concepts, and the chest-trained towers read and answer findings best in the grid but own only 7 of 36 chest cells. Ownership therefore belongs on the scorecard of the checkpoint to be deployed rather than on that of the vision tower it contains.

On chest radiographs, the label direction and the answer direction come apart. On natural images, the label direction and the model’s answer direction are substantially aligned, and writing the label direction controls the answer. On chest radiographs the two are nearly orthogonal: the label direction barely moves its own answer while moving the other findings’ answers, and the answer direction is owned in about half of the chest cells. The flat rows of the chest write matrices fit a reader that responds to a signal shared across findings rather than to each finding’s own direction (Pahde et al., 2025); which signal that is remains open. A legible label need not mark the direction along which the reader chooses its answer, so concept control is a property of a direction and a reader together.

What a steering evaluation should report. Probe accuracy and answer accuracy, the two quantities most often reported for medical VLMs, are both high in this grid, yet ownership is 13%. Random and sham writes show that a direction reaches the answer; only directions fitted the same way for the competing findings, written at the same locus and dose in the same patients, show whether the direction belongs to its label. Studies that use probe normals as interventions should report the write matrix W , compare O_q across layers, doses, and models, and pair every clinical write with a natural-image control, where the same procedure produces selective control.

7 Conclusion

Reading a clinical concept from a vision-language model’s visual representation does not make its probe direction a handle on that concept. Across 25 checkpoints and two chest-radiograph datasets, concepts are readable in 59% and answerable in 66% of cells but owned in 13%, because the concept’s own direction barely moves its answer while the directions fitted for the other findings move it as much

or more. Under the same procedure 75% of natural-object cells are owned, also at matched answer ability. Checkpoints that write identical vectors own different concepts, and the model’s own answer direction, nearly orthogonal to the probe direction on chest radiographs, provides the handle that the label direction lacks. Readability is a property of a classifier fitted to the representation; control is a property of a direction and the model that reads it.

8 Limitations

The grades describe linear writes at two loci, the consumed block and the connector output, at one primary dose with a five-point sweep; deeper language-model layers and nonlinear edits may behave differently. Each dataset contributes six concepts, and the chest labels are report-derived, with radiologist labels available for a CheXpert validation subset. The grade needs a discrete answer. MAIRA-2 (Bannur et al., 2024) emits reports rather than answers and fails the image-free mapping check on every template (Appendix A.3), and the discrete-diffusion report generator of Chen et al. (2026) has the same shape, so both fall outside the evaluation.

Ownership is a verdict on the write comparison rather than a predictor of effect size. On the four semantic endpoints, adding the ownership contrast to write magnitude, probe selectivity, and clean-answer AUROC adds no held-out predictive value beyond a permutation null ($p = 0.26$ and $p = 0.26$ under the concept and block splits; Appendix A.10). The answer direction is fitted on training rows to the model’s own clean answers and written on test rows and held-out templates, so it is a handle on the model’s answer. Its median AUROC against the report label is 0.66, below the probe’s 0.80, and answer control and clinical fidelity remain distinct objectives.

Reproducibility statement

The registered protocol file fixes the loci, projection, probe settings, dose grid, reference family, templates, modules, gates, and decision rules; its identifier and the evaluation-code version are recorded in every block’s scorecard. The release contains the evaluation code and adapters, the protocol file, the cohort manifests with their seeds, the fitted directions, the per-image outcomes of every module, every block’s scorecard and coverage file, the rendered prompts, and the pinned checkpoint revisions (Table 2); every table and figure in the paper is regenerated from those files by the scripts in the release. Appendix F states the consumed block of every family, the consumed-token mask, the projection and probe construction, the control sampling procedures, the cohort roles, and the protocol amendments with their dates.

Ethics statement

The study uses three public datasets under their published terms: NIH ChestX-ray14 (NIH Clinical Center), CheXpert Plus (Stanford AIMI, accessed through its data-use agreement), and COCO (annotations CC BY 4.0). All radiographs are de-identified by their providers; no protected health information was accessed, and no image or report is redistributed. The write test evaluates model behaviour under activation edits and is not a diagnostic tool; its grades describe what a probe direction does inside a model and carry no claim of clinical benefit.

References

- Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes. In *International Conference on Learning Representations Workshop*, 2017.
- Dana Arad, Aaron Mueller, and Yonatan Belinkov. SAEs are good for steering—if you select the right features. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 10241–10259, 2025.

- Shuai Bai, Yuxuan Cai, Ruizhe Chen, Keqin Chen, Xionghui Chen, et al. Qwen3-VL technical report. *arXiv preprint arXiv:2511.21631*, 2025a.
- Shuai Bai, Keqin Chen, Xuejing Liu, et al. Qwen2.5-VL technical report. *arXiv preprint arXiv:2502.13923*, 2025b.
- Shruthi Bannur, Kenza Bouzid, Daniel C. Castro, Anton Schwaighofer, Anja Thieme, Sam Bond-Taylor, Maximilian Ilse, Fernando Pérez-García, Valentina Salvatelli, Harshita Sharma, et al. MAIRA-2: Grounded radiology report generation. *arXiv preprint arXiv:2406.04449*, 2024.
- Yonatan Belinkov. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1), 2022.
- Nora Belrose, David Schneider-Joseph, Shauli Ravfogel, Ryan Cotterell, Edward Raff, and Stella Biderman. LEACE: Perfect linear concept erasure in closed form. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Pierre Chambon, Jean-Benoit Delbrouck, Thomas Sounack, Shih-Cheng Huang, Zhihong Chen, Maya Varma, Steven QH Truong, Chu The Chuong, and Curtis P. Langlotz. CheXpert Plus: Augmenting a large chest x-ray dataset with text radiology reports, patient demographics and additional image formats. *arXiv preprint arXiv:2405.19538*, 2024.
- Junying Chen, Chi Gui, Ruyi Ouyang, Anningzhe Gao, Shunian Chen, Guiming Hardy Chen, Xidong Wang, Ruifei Zhang, Zhenyang Cai, Ke Ji, et al. HuatuoGPT-Vision, towards injecting medical visual knowledge into multimodal LLMs at scale. *arXiv preprint arXiv:2406.19280*, 2024a.
- Lifeng Chen, Tianqi You, Hao Liu, Zhimin Bao, Jile Jiao, Xiao Han, Zhicai Ou, Tao Sun, Xiaofeng Mou, Xiaojie Jin, and Yi Xu. ECHO: Efficient chest x-ray report generation with one-step block diffusion. *arXiv preprint arXiv:2604.09450*, 2026.
- Zhihong Chen, Maya Varma, Justin Xu, Magdalini Paschali, Dave Van Veen, Andrew Johnston, Alaa Youssef, Louis Blankemeier, Christian Bluethgen, Stephan Altmayer, et al. A vision-language foundation model to enhance efficiency of chest X-ray interpretation. *arXiv preprint arXiv:2401.12208*, 2024b.
- Zehao Deng, Tianjie Ju, Zheng Wu, Liangbo He, Jun Lan, Huijia Zhu, Weiqiang Wang, and Zhuosheng Zhang. Causal probing for internal visual representations in multimodal large language models. *arXiv preprint arXiv:2605.05593*, 2026. Accepted at EMNLP 2026 Main.
- Yanai Elazar, Shauli Ravfogel, Alon Jacovi, and Yoav Goldberg. Amnesic probing: Behavioral explanation with amnesic counterfactuals. *Transactions of the Association for Computational Linguistics*, 9:160–175, 2021.
- Javier Ferrando, Enrique Lopez-Cuena, Pablo Agustin Martin-Torres, Daniel Hincos, Anna Arias-Duart, and Dario Garcia-Gasulla. Language models can explain visual features via steering. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.
- Atticus Geiger, Zhengxuan Wu, Christopher Potts, Thomas Icard, and Noah Goodman. Finding alignments between interpretable causal variables and distributed neural representations. In *Conference on Causal Learning and Reasoning (CLear)*, pages 160–187, 2024.
- Gemma Team. Gemma 3 technical report. *arXiv preprint arXiv:2503.19786*, 2025.
- Stefan Haufe, Frank Meinecke, Kai Görden, Sven Dähne, John-Dylan Haynes, Benjamin Blankertz, and Felix Bießmann. On the interpretation of weight vectors of linear models in multivariate neuroimaging. *NeuroImage*, 87:96–110, 2014.
- John Hewitt and Percy Liang. Designing and interpreting probes with control tasks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*, pages 2733–2743, 2019.

- Jeremy Irvin, Pranav Rajpurkar, Michael Ko, Yifan Yu, Silviana Ciurea-Ilcus, Chris Chute, Henrik Marklund, Behzad Haghgoo, Robyn Ball, Katie Shpanskaya, Jayne Seekins, David A. Mong, Safwan S. Halabi, Jesse K. Sandberg, Ricky Jones, David B. Larson, Curtis P. Langlotz, Bhavik N. Patel, Matthew P. Lungren, and Andrew Y. Ng. CheXpert: A large chest radiograph dataset with uncertainty labels and expert comparison. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 33, pages 590–597, 2019.
- Been Kim, Martin Wattenberg, Justin Gilmer, Carrie Cai, James Wexler, Fernanda Viegas, and Rory Sayres. Interpretability beyond feature attribution: Quantitative testing with concept activation vectors (TCAV). In *International Conference on Machine Learning (ICML)*, pages 2668–2677, 2018.
- LASA Team, Weiwen Xu, Hou Pong Chan, Long Li, Mahani Aljunied, Ruifeng Yuan, et al. Lingshu: A generalist foundation model for unified multimodal medical understanding and reasoning. *arXiv preprint arXiv:2506.07044*, 2025.
- Chunyuan Li, Cliff Wong, Sheng Zhang, Naoto Usuyama, Haotian Liu, Jianwei Yang, Tristan Naumann, Hoifung Poon, and Jianfeng Gao. LLaVA-Med: Training a large language-and-vision assistant for biomedicine in one day. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2023.
- Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C. Lawrence Zitnick. Microsoft COCO: Common objects in context. In *European Conference on Computer Vision (ECCV)*, pages 740–755, 2014.
- Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. Improved baselines with visual instruction tuning. In *IEEE Conference on Computer Vision and Pattern Recognition*, 2024.
- Samuel Marks and Max Tegmark. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. In *First Conference on Language Modeling (COLM)*, 2024.
- Meta. Llama 3.2-Vision model card. https://github.com/meta-llama/llama-models/blob/main/models/llama3_2/MODEL_CARD_VISION.md, 2024.
- Angus Nicolson, Lisa Schut, J. Alison Noble, and Yarin Gal. Explaining explainability: Recommendations for effective use of concept activation vectors. *Transactions on Machine Learning Research*, 2025.
- Frederik Pahde, Maximilian Dreyer, Leander Weber, Moritz Weckbecker, Christopher J. Anders, Thomas Wiegand, Wojciech Samek, and Sebastian Lapuschkin. Navigating neural space: Revisiting concept activation vectors to overcome directional divergence. In *International Conference on Learning Representations (ICLR)*, 2025.
- Nina Panickssery, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. Steering Llama 2 via contrastive activation addition. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024.
- Kiho Park, Yo Joong Choe, and Victor Veitch. The linear representation hypothesis and the geometry of large language models. In *International Conference on Machine Learning (ICML)*, pages 39643–39666, 2024.
- Nikollette Pedersen, Regitze Syddendal, Veronika Cheplygina, and Théo Sourget. Look what the probes dragged in! real-world chest x-ray shortcuts in MedCLIP. *arXiv preprint arXiv:2608.12086*, 2026.
- Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning (ICML)*, pages 8748–8763, 2021.

- Achyuta Rajaram, Sarah Schwettmann, Jacob Andreas, and Arthur Conmy. Line of sight: On linear representations in VLLMs. *arXiv preprint arXiv:2506.04706*, 2025.
- Shauli Ravfogel, Yanai Elazar, Hila Gonen, Michael Twiton, and Yoav Goldberg. Null it out: Guarding protected attributes by iterative nullspace projection. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 7237–7256, 2020.
- Abhilasha Ravichander, Yonatan Belinkov, and Eduard Hovy. Probing the probing paradigm: Does probing accuracy entail task relevance? In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics*, pages 3363–3377, 2021.
- Andrew Sellergren, Sahar Kazemzadeh, Tiam Jaroensri, Atilla Kiraly, Madeleine Traverse, Timo Kohlberger, Shawn Xu, Fayaz Jamil, Cían Hughes, Charles Lau, et al. MedGemma technical report. *arXiv preprint arXiv:2507.05201*, 2025.
- Michael Tschannen, Alexey Gritsenko, Xiao Wang, Muhammad Ferjad Naeem, Ibrahim Alabdulmohsin, Nikhil Parthasarathy, Talfan Evans, Lucas Beyer, Ye Xia, Basil Mustafa, Olivier Hénaff, Jeremiah Harmsen, Andreas Steiner, and Xiaohua Zhai. SigLIP 2: Multilingual vision-language encoders with improved semantic understanding, localization, and dense features. *arXiv preprint arXiv:2502.14786*, 2025.
- Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. Steering language models with activation engineering. *arXiv preprint arXiv:2308.10248*, 2023.
- Weiyun Wang, Zhangwei Gao, Lixin Gu, Hengjun Pu, Long Cui, Xingguang Wei, et al. InternVL3.5: Advancing open-source multimodal models in versatility, reasoning, and efficiency. *arXiv preprint arXiv:2508.18265*, 2025.
- Xiaosong Wang, Yifan Peng, Le Lu, Zhiyong Lu, Mohammadhadi Bagheri, and Ronald M. Summers. ChestX-ray8: Hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. In *IEEE Conference on Computer Vision and Pattern Recognition*, pages 2097–2106, 2017.
- Zhengxuan Wu, Aryaman Arora, Atticus Geiger, Zheng Wang, Jing Huang, Dan Jurafsky, Christopher D. Manning, and Christopher Potts. AxBench: Steering LLMs? even simple baselines outperform sparse autoencoders. In *International Conference on Machine Learning (ICML)*, pages 67035–67080, 2025.
- Juan Manuel Zambrano Chaves, Shih-Cheng Huang, Yanbo Xu, Hanwen Xu, Naoto Usuyama, Sheng Zhang, Fei Wang, Yujia Xie, Mahmoud Khademi, Ziyi Yang, et al. A clinically accessible small multimodal radiology model and evaluation metric for chest X-ray findings. *Nature Communications*, 16(1):3108, 2025.
- Xiaohua Zhai, Basil Mustafa, Alexander Kolesnikov, and Lucas Beyer. Sigmoid loss for language image pre-training. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.
- Xun Zhu, Fanbin Mo, Xi Chen, Kaili Zheng, Shaoshuai Yang, Yiming Shi, Jian Gao, Miao Li, and Ji Wu. Lost in the hype: Revealing and dissecting the performance degradation of medical multimodal large language models in image classification. *arXiv preprint arXiv:2604.08333*, 2026.
- Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J. Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, J. Zico Kolter, and Dan Hendrycks. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023.

A Campaign details

This appendix lists the checkpoints and their consumed loci, the acceptance decisions, the coverage of the campaign, and the per-concept results of every completed block.

A.1 Labels, registration, and artifact

Labels. We use each dataset’s supplied report-derived labels: NIH ChestX-ray14 finding labels (Wang et al., 2017), the CheXpert labeler output (Irvin et al., 2019) distributed with CheXpert Plus (Chambon et al., 2024), and COCO instance annotations (Lin et al., 2014). Rows are positive for label 1, negative for label 0, and *unknown* when the labeler output is uncertain or blank, which is the uncertainty convention of the CheXpert labeler (Irvin et al., 2019). We exclude unknown rows from probe fitting, selectivity, and answer AUROC. We write and score them like all other rows. Fixed manifests drawn with recorded seeds define the cohorts, one manifest per dataset for every block. No patient and no image appears in two roles of a dataset. Every cohort is disjoint from every pilot-study cohort. Table 3 reports per-concept counts for the three scored cohorts: the training cohort for fitting the scaler, probes, and answer direction; the 400-row calibration cohort for the readable and answerable grades; and the 600-row test cohort for every write. A second CheXpert cohort contains 200 frontal films from the CheXpert validation set, one per patient, with the radiologists’ consensus labels (Irvin et al., 2019). This cohort has zero patient overlap with any campaign cohort (Table 20).

Registration. We froze the protocol before scoring any grid block. The loci, projection and probe settings, dose, reference family, six templates, module set, seven gates, and three decision rules match the registered protocol file and its specified code version. We registered the pilot study’s crossover and confirmation analyses before running them (Appendix B). One analysis pass over the packaged outcomes produces all reported results for each block. After completing the grid, we added the leaderboard, write-flow share, and per-image examples as descriptive summaries of these outputs.

Artifact. The release includes evaluation code, adapters, the registered protocol file, cohort manifests with their seeds, and fitted directions. It provides per-image outcomes for every module (write, dose, refit, locus, and template rows), scorecards and coverage files for every block, and rendered prompts. Each new checkpoint requires one adapter that identifies its input block and passes the gates.

A.2 Grid and consumed loci

The towers include CLIP (Radford et al., 2021) in the LLaVA families, SigLIP (Zhai et al., 2023) in Gemma 3, and a SigLIP 2 design (Tschannen et al., 2025) in Qwen3-VL; MedGemma’s MedSigLIP tower is a medically tuned SigLIP released with that model (Sellersgren et al., 2025). The three towers trained on chest radiographs are the XraySigLIP tower of CheXagent-2-3B, the EVA-CLIP tower of CheXagent-8B, and the BiomedCLIP-CXR tower of LLaVA-Rad-7B; HuatuoGPT-Vision-7B tunes on medical data over the same stock CLIP tower as LLaVA-1.5. Consumed-token counts range from 128 to 4,096 per image, and each family uses a fixed image resolution. The Llama 3.2 Vision projector also reads five intermediate local-layer outputs that bypass the final global block. LLaVA computes the final CLIP block but consumes the penultimate block. At every checkpoint, Gate (C) verifies that the write reaches the consumer and the answer logits and changes nothing outside the consumed tokens.

A.3 Acceptance decisions

Seven checks run on the 16 preflight rows at both loci before a block is scored: (A) bitwise determinism; (B) identity at $\alpha = 0$; (C) reach, under which the write must change the consumer’s input and answer logits without changing anything outside the consumed tokens; (D) batched-versus-single agreement of the answer logits within 0.25; (E) image-free semantic mapping of every template; (F)

Table 2: **Checkpoints and families.** Per checkpoint: vision tower, consumed tokens per image, input size, eligible templates, and pinned Hugging Face revision. Bold family rows give mean selectivity and clean-answer AUROC over clinical cells and the owned clinical and COCO cells, shaded by owned share.

Checkpoint	Vision tower	Tokens	Image	Templates	Revision	$\bar{s}$	AUROC	owned cells	
								clinical	COCO
Qwen2.5-VL (Bai et al., 2025b)						0.132	0.635	5/48	24/24
Qwen2.5-VL-3B	native ViT 675M	576→144	336 ²	IYWYIAWA	6628554				
Qwen2.5-VL-7B	native ViT 675M	576→144	336 ²	all	cc59489				
Qwen2.5-VL-32B	native ViT 675M	576→144	336 ²	all	7cfb30d				
Qwen2.5-VL-72B	native ViT 675M	576→144	336 ²	all	89c8620				
Qwen3-VL (Bai et al., 2025a)						0.137	0.735	4/36	18/18
Qwen3-VL-4B	SigLIP2-based	≤1024	≤ 512 ²	all	ebb281e				
Qwen3-VL-8B	SigLIP2-based	≤1024	≤ 512 ²	all	0c351dd				
Qwen3-VL-32B	SigLIP2-based	≤1024	≤ 512 ²	all	0cfaf48				
InternVL3.5 (Wang et al., 2025)						0.125	0.807	5/36	17/18
InternVL3.5-8B	InternViT-300M	1024	448 ²	all	741a7d0				
InternVL3.5-14B	InternViT-300M	1024	448 ²	all	226b96d				
InternVL3.5-38B	InternViT-6B	1024	448 ²	all	7c830fc				
Gemma 3 (Gemma Team, 2025)						0.124	0.564	2/36	17/18
Gemma 3 4B	SigLIP-So400m	4096	896 ²	all	093f9f3				
Gemma 3 12B	SigLIP-So400m	4096	896 ²	all	96b6f1e				
Gemma 3 27B	SigLIP-So400m	4096	896 ²	all	005ad34				
MedGemma (medical) (Sellersgren et al., 2025)						0.164	0.835	2/24	10/12
MedGemma 4B	MedSigLIP	4096	896 ²	all	290cda5				
MedGemma 27B	MedSigLIP	4096	896 ²	all	2d3e00e				
Lingshu (medical) (LASA Team et al., 2025)						0.142	0.820	11/24	12/12
Lingshu 7B	Qwen2.5-VL ViT	576→144	336 ²	all	b98aecd				
Lingshu 32B	Qwen2.5-VL ViT	576→144	336 ²	all	36b9827				
LLaVA-1.5 (Liu et al., 2024)						0.124	0.516	0/24	7/12
LLaVA-1.5-7B	CLIP ViT-L/14-336	576	336 ²	IYWY	b234b80				
LLaVA-1.5-13B	CLIP ViT-L/14-336	576	336 ²	IYWYIAWA	5dda288				
LLaVA-Med (medical) (Li et al., 2023)						0.136	0.560	1/12	1/6
LLaVA-Med 1.5 7B	CLIP ViT-L/14-336	576	336 ²	IYWY	91bb16c				
Llama 3.2 Vision (Meta, 2024)						0.119	0.688	0/12	1/6
Llama 3.2 Vision 11B	ViT-H/14	1601	560 ²	IBWB	9eb2daa				
CheXagent (medical) (Chen et al., 2024b)						0.180	0.857	7/24	2/12
CheXagent-2-3B	XraySigLIP ViT-L/16	1024	512 ²	IYWYIAWA	8f19b53				
CheXagent-8B	EVA-CLIP ViT 1.4B	1025→128	448 ²	IYWYIAWA	4934e91				
LLaVA-Rad (medical) (Zambrano Chaves et al., 2025)						0.209	0.793	0/12	0/6
LLaVA-Rad-7B	BiomedCLIP-CXR-518	1369	518 ²	IYWY	dcd6c6c				
HuatuoGPT-Vision (medical) (Chen et al., 2024a)						0.136	0.746	1/12	4/6
HuatuoGPT-Vision-7B	CLIP ViT-L/14-336	576	336 ²	all	34dfcdb				

throughput; and (G) fp32 recomputation of the answer logits. Deviations in (D) are declared and neutralised by scoring every condition of a question in one fixed batch composition. Templates that fail (E) are ineligible; a checkpoint whose yes/no template fails is graded on its eligible B-positive template (IB for Llama 3.2 Vision), and its scorecards record the template used and every declared deviation.

All 75 completed blocks pass checks (A), (B), (C), and (G): determinism error is zero, change at $\alpha = 0$ is zero, and fp32-versus-model logit differences are below 0.13. In check (D), the Qwen3-VL, Gemma 3, MedGemma, InternVL, and Llama families exceed the 0.25-logit tolerance (up to 1.84 logits for Gemma 3 12B). We record these as declared deviations. Fixed batch composition neutralises them. Check (E) fails across all datasets for the A/B templates for LLaVA-1.5-7B and LLaVA-Med, the B-mapped templates for Qwen2.5-VL-3B and LLaVA-1.5-13B, and the yes/no and A-mapped templates for Llama 3.2 Vision 11B. We record affected cells as ineligible. Check (E) fails for every template of MAIRA-2 (Bannur et al., 2024) on NIH: the model answers “No” to every finding on all 16 preflight images and “A” to every forced choice under both letter mappings, so no template carries a graded answer and the block produces no outcomes. Two Gemma 3 checkpoints answer “yes” to most chest questions; Gemma 3 4B does so on 94–97% of rows (clean $P(\text{yes}) > 0.99$).

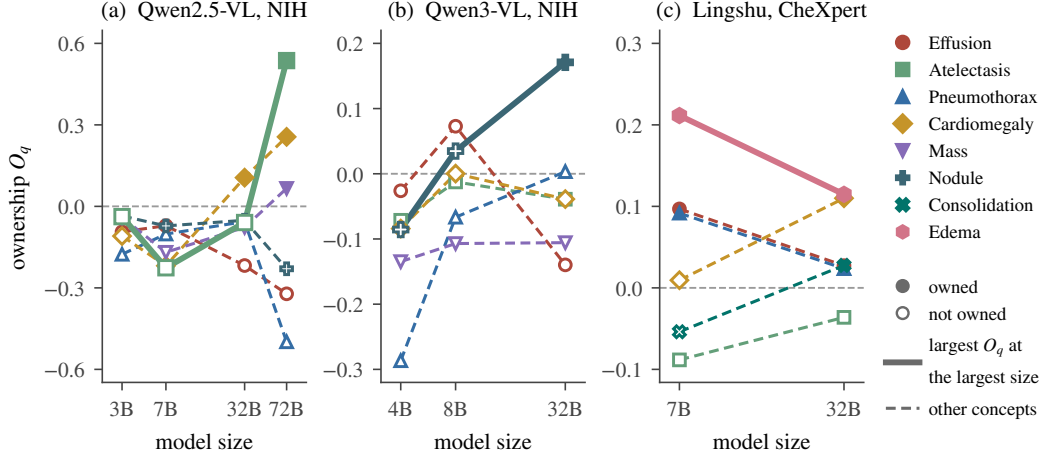


Figure 7: **Size ladders.** Ownership O_q per concept across sizes for Qwen2.5-VL and Qwen3-VL on NIH and Lingshu on CheXpert. Solid lines mark the concept with the largest ownership at the largest size (Atelectasis, Nodule, Edema), filled markers owned cells; vertical scales differ by panel.

Table 3: **Label counts per cohort and concept.** Positive / negative / unknown rows of the training, calibration, and test cohorts of the fixed manifests. Unknown means the CheXpert labeler was uncertain or blank.

Dataset	Concept	train (pos / neg / unk)			calibration (pos / neg / unk)			test (pos / neg / unk)		
NIH ChestX-ray14	Effusion	2423	15789	0	20	380	0	46	554	0
	Atelectasis	2311	15901	0	33	367	0	57	543	0
	Pneumothorax	1254	16958	0	13	387	0	17	583	0
	Cardiomegaly	881	17331	0	15	385	0	23	577	0
	Mass	1263	16949	0	15	385	0	32	568	0
	Nodule	1549	16663	0	27	373	0	47	553	0
CheXpert Plus	Effusion	5897	4277	9826	117	93	190	158	130	312
	Atelectasis	2891	86	17023	67	4	329	84	2	514
	Pneumothorax	1086	6456	12458	17	136	247	32	213	355
	Cardiomegaly	2312	1903	15785	49	46	305	71	58	471
	Consolidation	966	3652	15382	25	83	292	25	113	462
	Edema	3814	2231	13955	70	54	276	133	59	408
COCO	person	10877	9123	0	228	172	0	321	279	0
	dog	693	19307	0	10	390	0	15	585	0
	car	2051	17949	0	48	352	0	60	540	0
	chair	2183	17817	0	59	341	0	63	537	0
	bottle	1455	18545	0	34	366	0	47	553	0
	bicycle	567	19433	0	9	391	0	14	586	0

The answerability grade records this behaviour. Three checkpoints (LLaVA-1.5 7B/13B, LLaVA-Med) answer at chance.

A.4 Coverage and cost

Every checkpoint has a completed core write matrix and calibration pass on NIH ChestX-ray14, CheXpert Plus, and COCO (Table 4). The campaign targets seven modules per block: the write matrix, the calibration pass, the five additional templates, the dose sweep, the two probe refits, the connector locus, and the connector-locus calibration. 74 of 75 blocks carry all seven modules. A probe-graded block whose write matrix is ineligible or incomplete contributes readability alone. Every block’s scorecard also reports the signed dose range of O_q over $\alpha \in \{-0.5, \dots, +0.5\}$, the standard deviation of O_q over the three fit seeds (seeds 1 and 2 resample the training units with replacement and refit the scaler and the six probes), ownership and the absolute write effects at the

connector locus, and the wording (*is* versus *show*) and mapping (A-present versus B-present) contrasts of O_q across the six templates. Qwen2.5-VL-72B on CheXpert Plus is the partial block: its core write matrix, calibration, dose sweep, refits, and connector-locus calibration are complete; its prompt-template and connector-locus modules are not. Llama 3.2 Vision 11B is graded on its eligible B-positive template IB. The campaign used about 3,000 A100-hours. A chest block for a 7B checkpoint takes about ten GPU-hours.

Table 4: **Coverage.** Modules completed per block at packaging, from the campaign manifest. Brackets name the primary template when it is not the *is/yes-no* template.

Checkpoint	NIH ChestX-ray14	CheXpert Plus	COCO
Qwen2.5-VL-3B	C Cal P D R L Lc A Ad	C Cal P D R L Lc A Ad	C Cal P D R L Lc A Ad
Qwen2.5-VL-7B	C Cal P D R L Lc A Ad An E T Pr	C Cal P D R L Lc A Ad An E T	C Cal P D R L Lc A Ad An T Pr
Qwen2.5-VL-32B	C Cal P D R L Lc A Ad	C Cal P D R L Lc A Ad	C Cal P D R L Lc A Ad
Qwen2.5-VL-72B	C Cal P D R L Lc	C Cal D R Lc	C Cal P D R L Lc
Qwen3-VL-4B	C Cal P D R L Lc A Ad	C Cal P D R L Lc A Ad	C Cal P D R L Lc A Ad
Qwen3-VL-8B	C Cal P D R L Lc A Ad E T Pr	C Cal P D R L Lc A Ad	C Cal P D R L Lc A Ad T
Qwen3-VL-32B	C Cal P D R L Lc A Ad	C Cal P D R L Lc A Ad	C Cal P D R L Lc A Ad
InternVL3.5-8B	C Cal P D R L Lc A Ad E Pr	C Cal P D R L Lc A Ad	C Cal P D R L Lc A Ad
InternVL3.5-14B	C Cal P D R L Lc A Ad	C Cal P D R L Lc A Ad	C Cal P D R L Lc A Ad
InternVL3.5-38B	C Cal P D R L Lc A Ad	C Cal P D R L Lc A Ad	C Cal P D R L Lc A Ad
Gemma 3 4B	C Cal P D R L Lc A Ad Pr	C Cal P D R L Lc A Ad	C Cal P D R L Lc A Ad
Gemma 3 12B	C Cal P D R L Lc A Ad An E T Pr	C Cal P D R L Lc A Ad An E	C Cal P D R L Lc A Ad An
Gemma 3 27B	C Cal P D R L Lc A Ad Pr	C Cal P D R L Lc A Ad	C Cal P D R L Lc A Ad
MedGemma 4B	C Cal P D R L Lc A Ad Pr	C Cal P D R L Lc A Ad E	C Cal P D R L Lc A Ad
MedGemma 27B	C Cal P D R L Lc A Ad	C Cal P D R L Lc A Ad	C Cal P D R L Lc A Ad
Lingshu 7B	C Cal P D R L Lc A Ad	C Cal P D R L Lc A Ad	C Cal P D R L Lc A Ad
Lingshu 32B	C Cal P D R L Lc A Ad An E T Pr	C Cal P D R L Lc A Ad An E T	C Cal P D R L Lc A Ad An
LLaVA-1.5-7B	C Cal P D R L Lc A Ad	C Cal P D R L Lc A Ad	C Cal P D R L Lc A Ad
LLaVA-1.5-13B	C Cal P D R L Lc A Ad	C Cal P D R L Lc A Ad	C Cal P D R L Lc A Ad
LLaVA-Med 1.5 7B	C Cal P D R L Lc A Ad	C Cal P D R L Lc A Ad	C Cal P D R L Lc A Ad
CheXagent-2-3B	C Cal P D R L Lc	C Cal P D R L Lc	C Cal P D R L Lc
CheXagent-8B	C Cal P D R L Lc	C Cal P D R L Lc	C Cal P D R L Lc
LLaVA-Rad-7B	C Cal P D R L Lc	C Cal P D R L Lc	C Cal P D R L Lc
HuatuoGPT-Vision-7B	C Cal P D R L Lc	C Cal P D R L Lc	C Cal P D R L Lc
Llama 3.2 Vision 11B	C Cal P D R L Lc A [IB]	C Cal P D R L Lc A [IB]	C Cal P D R L Lc A [IB]

Legend: C = write matrix, Cal = calibration pass, P = five additional templates, D = dose sweep, R = two probe refits, L = connector locus, Lc = connector-locus calibration, A = alternative direction constructions, Ad = displacement lifts, An = answer direction, E = extended competitor family, T = token-weighted writes, Pr = fp32 and batch-size-one rescoring.

A.5 Ownership of every concept

Tables 5–7 report the ownership contrast of every concept cell, one table per dataset. Its cells are shaded by magnitude at the thresholds 0.02, 0.10, and 0.25, terracotta for positive and slate for negative ownership, and its checkpoints follow the order of Table 1. Figure 8 summarises the structure of the underlying write matrices. Figure 9 shows grades for individual images.

Table 5: **Ownership of every concept, NIH ChestX-ray14.** O_q at $\alpha = +0.25$: bold is owned, $\dagger$ a stronger competitor, $\ddagger$ unresolved; the last row counts owned cells. Shading darkens with $|O_q|$ (terracotta positive, blue negative).

Checkpoint	Effusion	Atelectasis	Pneumothorax	Cardiomegaly	Mass	Nodule
Lingshu 32B	-0.01	-0.08	+0.14	+0.05	-0.08	-0.18
CheXagent-2-3B	-0.00 $\ddagger$	+0.04	-0.00 $\dagger$	-0.01	+0.01	-0.01
Qwen2.5-VL-72B	-0.32	+0.54	-0.50	+0.26	+0.07	-0.23
Lingshu 7B	+0.02	+0.01	+0.01	+0.17	-0.09	+0.02
Qwen3-VL-4B	-0.03 $\dagger$	-0.07	-0.29	-0.08	-0.13	-0.09
Qwen3-VL-32B	-0.14	-0.04	+0.00 $\ddagger$	-0.04	-0.11	+0.17
InternVL3.5-38B	+0.00 $\ddagger$	-0.02	-0.00	-0.03	+0.02	+0.00
Gemma 3 12B	-0.83	-0.57	-0.17	-0.19	-0.65	+0.17
MedGemma 4B	+0.01	-0.56	+0.01	-0.07	-0.27	+0.09
InternVL3.5-14B	+0.01	-0.01	-0.00	-0.00	+0.00	-0.01
Qwen2.5-VL-32B	-0.22	-0.06	-0.06	+0.11	-0.08	-0.05
InternVL3.5-8B	+0.00	-0.02	-0.00	+0.00	-0.01	-0.01
HuatuoGPT-Vision-7B	-0.04	-0.03	-0.03	-0.08	-0.04	+0.01
LLaVA-Med 1.5 7B	-0.00	-0.00	-0.04	-0.01	-0.01	-0.02
CheXagent-8B	+0.01	-0.01	-0.00	-0.00	+0.00	-0.01
Qwen2.5-VL-3B	-0.09	-0.04	-0.18	-0.11	-0.04	-0.04
Qwen2.5-VL-7B	-0.07 $\dagger$	-0.23	-0.10	-0.22	-0.17	-0.07
Qwen3-VL-8B	+0.07	-0.01	-0.07	+0.00	-0.11	+0.03
Gemma 3 27B	-0.04	-0.03	-0.01	+0.00 $\ddagger$	-0.08	-0.02
Gemma 3 4B	-0.00	-0.00	-0.01	-0.24	-0.28	-0.12
LLaVA-1.5-7B	-0.03	-0.01	-0.08	-0.13	+0.02	-0.02
MedGemma 27B	-0.06	-0.45	-0.01	+0.00	-0.13	-0.20
LLaVA-1.5-13B	-0.01	-0.05	-0.04	-0.03	+0.01	-0.02
Llama 3.2 Vision 11B	-0.04	-0.07	-0.05	+0.02	-0.06	-0.01
LLaVA-Rad-7B	-0.02	-0.01	-0.00	+0.00	-0.01	-0.03
Owned	1	2	1	4	2	3

Table 6: Ownership of every concept, CheXpert Plus. As Table 5.

Checkpoint	Effusion	Atelectasis	Pneumothorax	Cardiomegaly	Consolidation	Edema
Lingshu 32B	+0.03	-0.04 [†]	+0.02	+0.11	+0.03	+0.11
CheXagent-2-3B	+0.03	-0.02	+0.01	+0.02	+0.01	-0.00 [‡]
Qwen2.5-VL-72B	-0.12	+0.09	+0.03	-0.11	-0.29	-0.13
Lingshu 7B	+0.10	-0.09	+0.09	+0.01 [‡]	-0.05	+0.21
Qwen3-VL-4B	-0.01 [†]	-0.02	-0.23	-0.01	+0.09	+0.09
Qwen3-VL-32B	-0.04	+0.00 [‡]	+0.00	-0.03	-0.12	+0.05
InternVL3.5-38B	+0.02	-0.01	-0.00	-0.00 [‡]	+0.01	-0.02
Gemma 3 12B	-0.83	-0.13	-0.71	-0.47	-0.26	+0.15
MedGemma 4B	-0.20	-0.08	-0.60	+0.04	-0.20	+0.10
InternVL3.5-14B	+0.01	-0.00	+0.00	-0.01	+0.01	-0.00
Qwen2.5-VL-32B	-0.07	-0.19	-0.12	-0.19	-0.05	-0.02
InternVL3.5-8B	-0.01	-0.01	+0.00	-0.00	+0.01	-0.00
HuatuoGPT-Vision-7B	-0.02	-0.02	-0.00	+0.01	-0.03	-0.01
LLaVA-Med 1.5 7B	+0.00	-0.00	-0.02	-0.03	-0.00 [†]	-0.02
CheXagent-8B	-0.02	-0.02	-0.02	-0.07	+0.04	-0.02
Qwen2.5-VL-3B	-0.18	-0.02	-0.04	-0.05	-0.03	-0.08
Qwen2.5-VL-7B	+0.01	-0.05	-0.10	-0.00	-0.15	-0.18
Qwen3-VL-8B	-0.07 [†]	-0.00 [†]	-0.10	-0.00	-0.04 [†]	-0.06
Gemma 3 27B	-0.03	-0.00	-0.01	-0.00	-0.03	+0.00 [‡]
Gemma 3 4B	-0.00	-0.00	-0.65	-0.02	-0.35	-0.58
LLaVA-1.5-7B	+0.00	-0.09	-0.10	+0.03	-0.03	-0.00
MedGemma 27B	-0.01	+0.00	-0.06	+0.03	-0.02	-0.28
LLaVA-1.5-13B	-0.00	-0.03	-0.01	+0.00	+0.01	-0.01
Llama 3.2 Vision 11B	-0.01	-0.05	-0.08	-0.07	-0.05	-0.01
LLaVA-Rad-7B	-0.04	-0.01	+0.00	-0.07	+0.02	-0.00
Owned	6	0	5	3	5	6

Table 7: **Ownership of every concept, COCO.** As Table 5.

Checkpoint	person	dog	car	chair	bottle	bicycle
Lingshu 32B	+0.29	+0.23	+0.08	+0.53	+0.34	+0.22
CheXagent-2-3B	+0.07	+0.07	+0.03	-0.01	+0.04	+0.00
Qwen2.5-VL-72B	+0.66	+0.44	+0.95	+0.89	+0.62	+0.40
Lingshu 7B	+0.39	+0.74	+0.70	+0.76	+0.64	+0.85
Qwen3-VL-4B	+0.46	+0.13	+0.33	+0.71	+0.34	+0.07
Qwen3-VL-32B	+0.13	+0.12	+0.11	+0.23	+0.24	+0.07
InternVL3.5-38B	+0.01	+0.00	+0.01	+0.02	+0.01	+0.01
Gemma 3 12B	+0.19	+0.76	+0.48	+0.45	+0.18	+0.61
MedGemma 4B	+0.35	+0.03	+0.43	+0.19	+0.20	+0.14
InternVL3.5-14B	+0.02	+0.01	+0.01	+0.04	+0.03	+0.00 [‡]
Qwen2.5-VL-32B	+0.44	+0.12	+0.27	+0.83	+0.41	+0.19
InternVL3.5-8B	+0.03	+0.01	+0.04	+0.04	+0.04	+0.02
HuatuoGPT-Vision-7B	+0.02	+0.00 [‡]	+0.00	+0.03	+0.03	+0.00 [‡]
LLaVA-Med 1.5 7B	+0.02	+0.01	+0.01	-0.02	+0.00	-0.01
CheXagent-8B	-0.00	+0.01	-0.01	-0.03	-0.13	-0.03
Qwen2.5-VL-3B	+0.46	+0.42	+0.53	+0.68	+0.61	+0.39
Qwen2.5-VL-7B	+0.48	+0.75	+0.61	+0.87	+0.51	+0.81
Qwen3-VL-8B	+0.42	+0.51	+0.74	+0.79	+0.59	+0.43
Gemma 3 27B	+0.24	+0.36	+0.09	+0.28	+0.15	+0.75
Gemma 3 4B	+0.16	+0.54	+0.24	+0.34	+0.44	-0.27
LLaVA-1.5-7B	+0.06	+0.00	+0.01	+0.07	+0.01	+0.00
MedGemma 27B	+0.33	-0.04	+0.08	+0.18	+0.21	-0.04
LLaVA-1.5-13B	+0.05	+0.00 [‡]	+0.02	+0.08	+0.02	-0.03
Llama 3.2 Vision 11B	-0.03	-0.02	+0.02	-0.02	-0.04	+0.01
LLaVA-Rad-7B	-0.06	+0.01	-0.02	-0.06	-0.04	-0.00
Owned	21	19	19	20	18	16

A.6 Controls

Table 9 summarises the reference family and the per-block controls of Section 5.7 across the grid.

A.7 What each analysis scores its writes against

Table 10 lists the 20 graded analyses (5,844 cells); the packaged per-block statistics populate each row and check its declared reference and sham. The write matrix scores its own write against the 95th percentile of the 119-direction random family and the absolute sham, decides by simultaneous max- T bounds, and permutes the direction it writes. 18 of the analyses share this reference, 19 this rule, and 16 this sham. The semantic endpoints instead take the maximum of an 18-direction random family of their own, and the answer direction under the held-out templates has the random 95th percentile for 80 of its 270 (cell, template) pairs and the sham alone for the rest. The logit-scale grade decides by a percentile interval on the ownership contrast, whose maximum it recomputes in every draw. The probe refits, alternative constructions, displacement lifts, and expert-label refits (3,060 cells) permute the seed-0 logistic normal rather than the direction they write, and the token-weighted writes are referenced against the sham of the uniformly written normal.

A.8 Robustness of the deficit

The projected features are $x = R^\top \mathbf{h}$, where $R \in \mathbb{R}^{D \times 512}$ is the fixed Gaussian projection. Two lifts map vectors from the projected, train-scaled space back to the consumed space. A classifier coefficient β lifts as $\hat{\mathbf{w}} \propto R \text{diag}(1/s)\beta$. The resulting written direction scores the concept as the probe does (Equation 2). A displacement $\mathbf{u}$ lifts via its minimum-norm preimage as $\hat{\mathbf{w}} \propto R(R^\top R)^{-1} \text{diag}(s)\mathbf{u}$. This lift reproduces the displacement exactly: $R^\top \hat{\mathbf{w}} \propto \text{diag}(s)\mathbf{u}$. The lifts differ because $R^\top R$ is not a multiple of the identity. Measurements on every block with the displacement lifts show that $R^\top R/D$ is full rank, with eigenvalues 1.7×10^{-4} – 5.7×10^{-3} and condition number 5.3–33.2. We compute the difference-of-means and pattern vectors in the same projected, train-scaled space as the logistic normal and report both lifts. The alternative-construction module scores the coefficient-lift family. The displacement lift gives the geometrically matched comparison for these two families: a difference of class means and a Haufe pattern are displacements in the projected space rather than classifier coefficients. We form the orthogonalised normal by removing the span of the other five normals from the logistic normal. We refit the residualised normal on features residualised against the other five labels.

Alternative directions, weights, and competitor families. Token-weighted variants multiply the per-token dose $\alpha \|\mathbf{h}_{it}\|$ by a weight w_t whose mean over the consumed tokens is one. The softmax variant weights tokens by the softmax of their probe scores. It preserves the uniform write’s L1 dose. The top-quarter variant ($w_t = 4$ on the 25% highest-scoring tokens) has twice the uniform write’s Frobenius norm. The extended competitor family adds a direction for every other dataset label with at least 100 known positives and 100 known negatives (NIH: Consolidation, Edema, Infiltration; CheXpert: Enlarged Cardiomeastinum, Fracture, Lung Lesion, Lung Opacity, Pneumonia, Support Devices). We fit and write each added direction like the protocol normals. Under this family, a cell is owned when its write meets the steering reference and holds the simultaneous advantage over the five protocol competitors and every extra direction. Table 15 reports every construction, lift, token weighting, and competitor family per dataset. Each panel includes the protocol’s logistic normal on the same blocks.

Non-clinical attributes. The attribute directions use three labels from each dataset’s metadata: the view field (anteroposterior or portable against posteroanterior), recorded sex, and age at least 60. We fit these directions using the same images, features, projection, and probe settings as the clinical probes. We then lift and write them with the six clinical directions as one nine-direction family. Attribute and clinical questions share the same steering reference in all 38 blocks of Table 17: the 119-direction random family written at the same dose and the coordinate-permutation sham. We

Table 8: **Steering reference against verdict.** Primary write-matrix cells of the included blocks, cross-classified by the steering reference and by the simultaneous-bound verdict against the five other clinical directions.

Cells	Steering reference	advantage	stronger competitor	unresolved	total
Chest, all cells	met	38	9	9	56
	not met	24	187	33	244
Chest, readable and answerable	met	28	5	5	38
	not met	13	74	16	103
COCO, all cells	met	113	0	4	117
	not met	10	17	6	33

Table 9: **Reference family and controls.** Median and interquartile range per dataset, over all cells in the first four rows and over blocks with the module in the next six; the last three rows count cells.

Quantity	NIH ChestX-ray14		CheXpert Plus		COCO (control)	
	median	IQR	median	IQR	median	IQR
Random-family p95 of W	+0.063	[+0.021, +0.165]	+0.066	[+0.027, +0.157]	+0.023	[+0.008, +0.055]
Sham write	+0.028	[+0.009, +0.075]	+0.027	[+0.008, +0.076]	+0.011	[+0.003, +0.034]
Concept write $W_{q,q}$	+0.010	[-0.006, +0.067]	+0.020	[-0.001, +0.082]	+0.127	[+0.020, +0.482]
Ownership O_q	-0.017	[-0.078, +0.000]	-0.014	[-0.065, +0.001]	+0.102	[+0.009, +0.433]
Signed dose range of O_q	+0.266	[+0.061, +0.444]	+0.162	[+0.034, +0.359]	+0.399	[+0.050, +0.612]
Refit SD of O_q (3 seeds)	+0.039	[+0.015, +0.072]	+0.037	[+0.017, +0.052]	+0.043	[+0.006, +0.074]
Connector-locus median O_q	-0.013	[-0.023, -0.005]	-0.009	[-0.020, -0.004]	+0.003	[+0.001, +0.007]
Label gap (logits)	-0.1	[-0.9, -0.0]	-0.3	[-1.3, +0.0]	-5.3	[-14.8, -0.3]
Wording contrast (IY–WY)	+0.033	[+0.008, +0.078]	+0.021	[+0.007, +0.035]	+0.028	[+0.007, +0.065]
Mapping contrast (IA–IB)	+0.037	[+0.008, +0.073]	+0.019	[+0.003, +0.055]	+0.030	[+0.007, +0.051]
Reference met	20 of 150		36 of 150		117 of 150	
Owned	13 of 150		25 of 150		113 of 150	
Reference met but not owned	7 of 150		11 of 150		4 of 150	

score each attribute question in 3 phrasings on the calibration rows. The phrasing used for the write reaches a median clean-answer AUROC of 0.679, compared with 0.716 for the phrasing in the same cell with the highest one-sided lower bound. Across the 114 attribute cells, the median gap is 0.009 and the largest gap is 0.275. 100 of those cells clear the answerable rule as written. We assess the comparison three ways: over every cell, over answerable cells alone, and over attribute cells paired with clinical cells in the same block within 0.05 clean-answer AUROC. This matching places 79 of the 114 attribute cells into 188 pairs and leaves 35 without a partner in their own block. Attribute probes reach a median AUROC of 0.896–0.999 against the attribute label and 0.63–0.74 against the model’s own clean answer. We summarise each attribute–finding pair by its median over blocks; over the 24 pairs the largest absolute raw model-space cosine between an attribute direction and a clinical normal is 0.11, for AP (portable) projection with Edema. Across the 38 blocks with the module, ownership counts are 12 of 114 cells for attributes and 23 of 228 for findings in the same family. The corresponding attribute and finding counts are 5/57 and 17/114 on CheXpert, and 7/57 and 6/114 on NIH.

Answer direction: rescue, loss, and templates. We lift the answer direction a_q of Section 5.6 as we lift the probe normal. We write it at the same dose, against the write matrix’s own 119-direction random family and a coordinate-permutation sham of the a_q under test, using the other five a_d as clinical competitors. A concept is *kept* when both a_q and the label direction $\hat{w}_q$ are owned, *rescued* when only a_q is owned, and *lost* when only $\hat{w}_q$ is owned (Table 16, top). The release reports both ownership contrasts, column selectivity, and the whitened cosine for every cell. We also write the a_q fitted on the primary template IY unchanged under the rewording WY and the answer-mapped templates IA, IB, WA, and WB. Each template uses its own clean baseline, the a_d of the other five concepts, and a coordinate-permutation sham of the a_q under test. The held-out templates score no random family of their own. The random 95th percentile is available where the additional-template

Table 10: **What each analysis scores its writes against.** Per graded analysis: scored rows, steering reference, decision rule, the direction its sham permutes, and graded (question, arm) cells.

Analysis	Rows	Reference	Rule	Sham	Cells
Write matrix, primary template	600 test	119 random p_{95} , sham	simultaneous max- T , 30 contrasts	the direction written	450
Probe refits, seeds 1 and 2	600 test	119 random p_{95} , sham, the write matrix's	simultaneous max- T , 30 contrasts	the seed-0 normal	900
Logit-scale grade	600 test	119 random p_{95} , sham, on the margin scale	percentile interval on O_q^m	the direction written	450
Unsaturated-row grade	unsaturated test	119 random p_{95} , sham, on those rows	simultaneous max- T , 30 contrasts	the direction written	450
Alternative direction constructions	600 test	119 random p_{95} , sham, the write matrix's	simultaneous max- T , 30 contrasts	the seed-0 normal	1,440
Displacement lifts	600 test	119 random p_{95} , sham, the write matrix's	simultaneous max- T , 30 contrasts	the seed-0 normal	684
Extended competitor family	600 test	119 random p_{95} , sham	simultaneous max- T , 8 or 11 contrasts	the direction written	54
Token-weighted writes	600 test	119 random p_{95} , sham, the sham unweighted	simultaneous max- T , 30 contrasts	the direction written	96
fp32 and batch-size-one rescoring	200 test	119 random p_{95} , sham	simultaneous max- T , 30 contrasts	the direction written	108
Answer direction, primary template	600 test	119 random p_{95} , sham	simultaneous max- T , 30 contrasts	the direction written	54
Answer direction, held-out templates	600 test	119 random p_{95} , sham in 80 pairs; sham alone in 190	simultaneous max- T , 30 contrasts	the direction written	270
Non-clinical attributes	600 test	119 random p_{95} , sham	simultaneous max- T , 72 contrasts	the direction written	114
Findings in the attribute family	600 test	119 random p_{95} , sham	simultaneous max- T , 72 contrasts	the direction written	228
Radiologist-label regrade	200 valid	119 random p_{95} , sham	simultaneous max- T , 30 contrasts	the direction written	114
Expert-label refits	600 test	119 random p_{95} , sham, the write matrix's	simultaneous max- T , 30 contrasts	the seed-0 normal	36
Projection seeds 1 and 2	600 test	119 random p_{95} , sham, that projection's	simultaneous max- T , 30 contrasts	the direction written	36
Crossed tower and reader	200 test	119 random p_{95} , sham, of the tower in use	simultaneous max- T , 30 contrasts	the direction written	132
Identical-tensor replay	200 test	119 random p_{95} , sham, the source block's	simultaneous max- T , 30 contrasts	the direction written	48
Semantic endpoints	600 test	18 random, maximum; sham, the endpoint's own	simultaneous max- T , 30 contrasts	the direction written	144
Fine-grained object family	600 test	119 random p_{95} , sham, the family's own	simultaneous max- T , 30 contrasts	the direction written	36

module scored one under that template, which is 80 of the 270 (cell, template) pairs; the remaining 190 pairs meet the steering reference against their own sham alone. Every pair takes the same 6×5 max- T verdict (Table 10). A block enters when every held-out template is scored on all 600 test rows (Table 16, bottom).

Direction and label geometry. Neither correlations among the six directions nor co-occurrence of the six report labels explains why a competing direction wins (Table 11). The direction cosine is the raw model-space cosine between two of a block's six write directions, averaged over all blocks' off-diagonal pairs. The label ϕ measures the correlation between two of the six test labels. On NIH, the six directions are close to orthogonal: their mean absolute off-diagonal cosine is 0.093, against 0.024 between pairs of random unit directions, and their signed mean is 0.019. The labels are nearly uncorrelated (mean $\phi = 0.026$). Yet NIH shows the same off-diagonal dominance as CheXpert, where directions and labels are correlated (mean absolute off-diagonal cosine 0.22, signed mean 0.21; Effusion–Consolidation 0.46 averaged over blocks, $\phi = 0.74$). The ownership contrast is positive, $O_q > 0$, in 25% of the 150 NIH cells and 29% of the 150 CheXpert cells. Each cell has five competing directions, so its strongest competitor matches any fixed competitor in one cell in

five by chance. The strongest competitor matches the most similar direction at this rate (21–22% of chest cells per dataset). It also corresponds to the label with the largest ϕ or the most co-occurring label at this rate (16–24%). Across the 1,500 off-diagonal chest cells, the competitor’s advantage $W_{q,d} - W_{q,q}$ is uncorrelated with the direction cosine (Spearman over cells 0.037, $p = 0.15$) and with the label ϕ (-0.001). At least two competitors exceed the own write in 163 of 300 chest cells (54%). Removing the strongest competitor therefore leaves O_q negative in those cells. The last three rows of Table 11 count the sign of O_q after removal, not the owned grade. On COCO, we restrict each cell’s competitors to a co-occurring family of objects (person with bicycle, $\phi = 0.14$; person with dog, $\phi = -0.06$; chair with bottle, $\phi = 0.12$; car with bicycle, $\phi = 0.13$; person with chair and bottle, mean $\phi = 0.08$). This leaves $O_q > 0$ in 91–96% of each family’s cells, compared with 82–90% of the same cells with all five competitors. Of the five directions in the Effusion case, Nodule is the *least* similar to Effusion (raw model-space cosine -0.06 in this block). It still moves the Effusion answer more than Effusion’s own direction (0.262 against 0.190). The most similar direction, Atelectasis (raw model-space cosine 0.49), moves it by 0.083.

Scale, saturation, refits, and numerics. The grades hold on the logit scale. Recomputing the grade from the margin $\ell_{\text{yes}} - \ell_{\text{no}}$ reproduces the probability-scale owned grade in 440 of 450 cells. Every owned cell retains a positive ownership contrast on that scale, $O_q^m > 0$. The full owned grade persists in 148 of the 151 owned cells, with the steering reference recomputed on the margin scale and a positive percentile interval for O_q^m (Table 12). Saturation counters a ceiling explanation: COCO blocks contain far more rows at $P(\text{yes}) > 0.99$ or < 0.01 than chest blocks (median block share 0.71 against 0.16 and 0.10). Restricting each cell to its unsaturated rows preserves $O_q > 0$ in all 38 owned chest cells and in 41 of the 252 others. Full regrading uses a steering reference recomputed on these rows and the simultaneous max- T advantage over the same six-direction family. It preserves the owned grade in 36 of 38 owned chest cells and raises 0 of the 252 others to owned. The same pattern holds in the clean-No, clean-Yes, label-present, and label-absent strata. Refitting the probes on resampled training patients, and grading each refitted direction against the seed-0 random 95th percentile and the seed-0 sham of the same block and question, changes O_q by a median of 0.02 (Table 10); the last column of Table 13 regrades seed 0 with the same draws and counts agreement with the paper’s grade. Rescoring the write grid on the first 200 test rows of each block, in fp32 and at batch size one, moves the 6×126 written cells of a block by at most 0.056. Table 12 gives that largest change and the largest in a clinical contrast $C_{q,d}$, the difference between the own write and a competing write on the same question. Regraded under either setting, verdicts change in 1 and steering references in 0 of the 108 grades, and the two point verdicts change in 0 (Table 12). The bf16 batch effects declared at the gates (candidate-logit differences of 0.00–1.84) affect no owned cell. The margin sign agrees in 70 of 76 blocks; the 6 disagreeing blocks contain no owned cells. Re-scoring the clean baseline in a second module gives bitwise agreement in 100% of rows in 44 of 74 blocks and in at least 0% of rows in the rest. Differences satisfying $|\Delta P(\text{yes})| > 0.1$ occur in at most 7.7% of rows. Writes at the connector locus are an order of magnitude smaller than those at the consumed block (median $|W_{q,q}|$ 0.008 against 0.032 on NIH and 0.008 against 0.127 on COCO). The readable and answerable grades use the 400 calibration rows for support. 25 of the 300 chest cells fail that cohort’s requirement of ten known positives and ten known negatives: CheXpert Atelectasis in each of the 25 CheXpert blocks, with 67 known positives and 4 known negatives among the 400 calibration rows. Every NIH concept meets the requirement on that cohort; Pneumothorax has the smallest count, at 13 positives. On COCO, bicycle is ineligible in each of the 25 COCO blocks, with 9 positives. Support counts for these two grades come solely from calibration. The test cohort carrying the writes has its own counts (84 positives and 2 negatives for CheXpert Atelectasis, 14 positives for COCO bicycle). Among eligible cells, 178 of 275 are readable (65%) and 199 of 275 are answerable (72%).

The answer direction and the label source. We fit the answer direction of Section 5.6 to one quantity: the model’s own clean answer margin $\ell_{\text{yes}} - \ell_{\text{no}}$ on the block’s primary template, read from clean forward passes with the write disabled. We use the block’s own projected, train-scaled features from the first 3,000 training-cohort rows in the cohort’s fixed manifest order. A seeded shuffle

determines this order on NIH, and a hash of patient or image identity determines it on CheXpert and COCO. Neither labels nor answers enter the ordering. For each question, we fit the margin from these features by ridge regression. We select the penalty by five-fold cross-validated R^2 over $\{0.1, 1, 10, 100\}$, then refit on all 3,000 rows with the chosen penalty. Across the 54 fitted questions, the selected penalty is 1 for 25, 10 for 28, and 100 for 1. The folds split rows: one per patient on CheXpert, one per image on COCO, and rows from 1,871 patients on NIH. We lift and normalise the fitted coefficient vector exactly as we do a probe normal. The radiologist-label cohort provides two distinct tests. First, we re-grade the unchanged directions fitted on report-derived labels on 200 films labelled by radiologist consensus. These patients appear in no campaign cohort (Table 20). The absolute difference $|O_q(\text{valid}) - O_q(\text{test})|$ has a 90th percentile of 0.021 over all cells. On the 76 cells with at least ten known positives and ten known negatives on both cohorts, readability agrees in 73 of 76 and answerability in 60 of 76. Second, we refit the six directions on those radiologist labels using the block’s own projection, its train-only scaler, and the protocol’s probe settings. We cross-fit over patients in 5 folds, scoring every film with a direction not fitted on it. We write the refitted directions on the 600 test rows at the primary dose and template, against the write matrix’s own 119-direction random family and its seed-0 sham, and take the same 6×5 max- T verdict; the sham here is the permutation of the report-label direction rather than of the refitted one (Table 10). Across the 6 CheXpert blocks with this refit, the expert-label and report-label directions have a median raw model-space cosine of 0.12 and a median covariance-whitened cosine of 0.56. The expert-label refit achieves a median held-out AUROC of 0.69 on the radiologist labels, against 0.79 for the report-label direction on the same films. Those two directions differ in two ways at once: the labels they were fitted to, and the number of rows available to fit them (160 per fold against a median of 5332 labelled training rows per concept). Three further arms separate the two, all scored on the same films, with the block’s own projection, its stored train-only scaler, the protocol’s probe settings and the same 5-fold cross-fitting over patients (Table 20, lower panel). Report-derived labels of those same 200 films reach 0.67; report-derived labels of a training subsample drawn to leave each concept the same 160 fitting rows as the refit, redrawn 20 times, reach 0.70; report-derived labels of every training row reach 0.79. Holding the number of fitting rows fixed, the label source moves the result by -0.01 ; holding the label source fixed, the sample size moves it by $+0.09$. The size-matched report refit also sits at a raw model-space cosine of 0.22 to the shipped direction and a whitened cosine of 0.69, against 0.12 and 0.56 for the expert refit. The distance between the refit and the shipped direction is the estimation sample, not the label source. The expert-label directions are owned in 5 of 36 cells, against 12 for the report-label directions of the same blocks.

Table 11 reports direction and label geometry. Table 12 reports scale, saturation, refit, connector, batch, and numerics statistics supporting Section 5.7. Table 13 reports the full grade under refitted probes. Table 14 reports paired shared-tower differences supporting Section 5.4. Table 15 reports alternative directions, lifts, token weights, and competitor families. Table 16 reports the answer direction on the primary and held-out templates. Table 17 reports non-clinical attribute directions. Table 19 reports what the answer direction reads and the column selectivity of W . Table 20 reports the grade on radiologist labels and separates the label source of a refitted direction from the number of rows it was fitted on. Table 21 reports the crossed tower and reader. Table 22 reports the four semantic endpoints. Table 23 reports the fine-grained object family with the difficulty-matched comparison (Appendix A.10).

A.9 Further results

This subsection reports the detail behind Sections 5.1–5.7.

Reading and answering. Median probe selectivity over the chest cells is 0.14, with quartiles 0.08 and 0.22; of the 199 answerable chest cells, 141 are also readable. Effusion and Cardiomegaly are the most readable of the four concepts both chest sets share, with mean selectivity 0.18 and 0.19 over 50 chest blocks. Table 8 cross-classifies every chest and COCO cell by the steering reference and the verdict.

Table 11: **Direction and label geometry against off-diagonal dominance.** One column per dataset, with the two chest sets pooled in the third. A “–” marks a row that is defined per dataset only.

	NIH ChestX-ray14	CheXpert Plus	both chest sets	COCO
<i>Geometry of the six directions and the six labels</i>				
mean off-diagonal cosine between the directions	0.019	0.215	–	0.043
mean off-diagonal $ \cos $ between the directions	0.093	0.218	–	0.058
mean $ \cos $ between random unit directions	0.024	0.024	–	0.024
mean off-diagonal ϕ between the labels	0.026	0.381	–	0.022
<i>Cells whose strongest competitor is (chance: one in five)</i>				
the most similar direction	31/150	33/150	64/300	31/150
the label with the largest ϕ	35/150	36/150	71/300	29/150
the most co-occurring label	24/150	32/150	56/300	22/150
<i>Spearman ρ over off-diagonal cells of $W_{q,d} - W_{q,q}$ with</i>				
the cosine between the two directions	0.085	0.048	0.037	0.042
the ϕ between the two labels	0.036	0.037	-0.001	0.007
<i>Sign of O_q</i>				
cells with $O_q > 0$	38/150	43/150	81/300	129/150
cells with $O_q > 0$ without the strongest competitor	62/150	75/150	137/300	136/150
cells with at least two competitors above the own write	88/150	75/150	163/300	14/150

Table 12: **Robustness of the owned grade.** Owned cells per dataset under each alternative scale, stratum, refit, and locus, beside the grade on the probability (p) and logit-margin (m) scales; the lower panel rescores in fp32 at batch size one.

Dataset	owned cells			unsaturated, $O_q > 0$		refits		median $ W_{q,q} $		connector
	p	m	both	owned	other	med. $ \Delta O $	stable	primary	connector	> p95
NIH ChestX-ray14	13/150	12/150	11	13/13	25/131	0.020	8/13	0.032	0.008	15/150
CheXpert Plus	25/150	27/150	25	25/25	16/121	0.023	10/25	0.032	0.010	16/144
COCO	113/150	116/150	112	105/106	16/37	0.020	107/113	0.127	0.008	106/150

Batch effects at the gates (61 blocks): batched-vs-single candidate-logit difference 0.00–1.84 (median 0.28); margin sign agreement in 55 of 61 blocks; owned cells in disagreeing blocks: 0.

Numerics: the same grid rescored in fp32 and at batch size one

Checkpoint	Dataset	fp32 weights and forward		bf16, batch size one	
		max $ \Delta W $	max $ \Delta C $	max $ \Delta W $	max $ \Delta C $
Qwen2.5-VL-7B	NIH ChestX-ray14	0.0087	0.0063	0.0065	0.0049
Qwen2.5-VL-7B	COCO	0.0060	0.0032	0.0028	0.0020
Qwen3-VL-8B	NIH ChestX-ray14	0.0177	0.0089	0.0204	0.0089
InternVL3.5-8B	NIH ChestX-ray14	0.0018	0.0014	0.0018	0.0014
Gemma 3 4B	NIH ChestX-ray14	0.0544	0.0493	0.0261	0.0377
Gemma 3 12B	NIH ChestX-ray14	0.0558	0.0359	0.0111	0.0109
Gemma 3 27B	NIH ChestX-ray14	0.0413	0.0186	0.0149	0.0053
MedGemma 4B	NIH ChestX-ray14	0.0125	0.0069	0.0056	0.0054
Lingshu 32B	NIH ChestX-ray14	0.0032	0.0023	0.0009	0.0007

Natural objects. Ownership margins on COCO span 0.06–0.96 in the Qwen, Lingshu, and Gemma families. The dose sweep ran on 25 NIH and 25 COCO blocks. Within a checkpoint, the counts of owned fine-grained and easy objects differ by -2 to 0 of six, and median probe selectivity is 0.058 for the fine-grained objects against 0.065 for the easy ones. Table 23 gives the remaining difficulty bins, the per-dataset counts, and the matched estimator.

Shared-tower pairs. Table 14 records the bitwise relation of every pair. The median paired difference in ownership is 0.6–4.6 times the refit standard deviation, up to 17.3 times for single concepts, and it exceeds that standard deviation in 11 of 12 pairs. On NIH Effusion, the identical write yields an O_q difference of $+0.79$ [$+0.76, +0.82$] between Gemma 3 12B and 27B. In 12B, competing directions win the chest answers outright ($O_{\text{Effusion}} = -0.83$ on both chest sets, $O_{\text{Mass}} = -0.65$ on NIH,

Table 13: **The full grade under refitted probes.** Cells per seed-0 grade (owned, stronger competitor, unresolved, fixed-family advantage without the reference) and how many keep it under both refits, one, or none.

Dataset	blocks	cells	owned				stronger competitor			unresolved			adv. no ref.		seed 0
			<i>n</i>	both	≥ 1	none	<i>n</i>	both	≥ 1	<i>n</i>	both	≥ 1	<i>n</i>	both	agrees
NIH ChestX-ray14	25	150	13	4	8	5	101	69	97	21	5	12	15	0	150/150
CheXpert Plus	25	150	25	6	17	8	95	63	89	21	3	10	9	3	150/150
COCO	25	150	113	99	110	3	17	13	17	10	4	7	10	2	149/150
<i>chest</i>	50	300	38	10	25	13	196	132	186	42	8	22	24	3	300/300
<i>all</i>	75	450	151	109	135	16	213	145	203	52	12	29	34	5	449/450

Table 14: **Same write, different reader: paired differences.** $\Delta O_q = O_q(\text{larger}) - O_q(\text{smaller})$ per shared-tower pair on the same 600 test rows, with paired patient-bootstrap 95% intervals; the last two columns divide $|\Delta O_q|$ by the two blocks' mean refit standard deviation.

Pair	Dataset	relation	ΔO_q CI	$ \Delta O_q $		ratio to refit SD	
			$\neq 0$	median	max	median	max
Gemma 3 4B $\rightarrow$ Gemma 3 12B	NIH ChestX-ray14	bit-identical	5/6	0.33	0.83	2.7	6.9
Gemma 3 4B $\rightarrow$ Gemma 3 12B	CheXpert Plus	bit-identical	6/6	0.29	0.83	2.1	5.9
Gemma 3 4B $\rightarrow$ Gemma 3 12B	COCO	bit-identical	5/6	0.23	0.87	1.6	6.0
Gemma 3 12B $\rightarrow$ Gemma 3 27B	NIH ChestX-ray14	bf16-equal	6/6	0.37	0.79	3.7	8.0
Gemma 3 12B $\rightarrow$ Gemma 3 27B	CheXpert Plus	bf16-equal	6/6	0.35	0.81	1.4	3.3
Gemma 3 12B $\rightarrow$ Gemma 3 27B	COCO	bf16-equal	5/6	0.15	0.40	1.7	4.3
Gemma 3 4B $\rightarrow$ Gemma 3 27B	NIH ChestX-ray14	bf16-equal	5/6	0.07	0.24	1.8	6.0
Gemma 3 4B $\rightarrow$ Gemma 3 27B	CheXpert Plus	bf16-equal	6/6	0.17	0.65	4.6	17.3
Gemma 3 4B $\rightarrow$ Gemma 3 27B	COCO	bf16-equal	6/6	0.16	1.02	1.4	8.8
MedGemma 4B $\rightarrow$ MedGemma 27B	NIH ChestX-ray14	bf16-equal	5/6	0.09	0.30	1.5	5.0
MedGemma 4B $\rightarrow$ MedGemma 27B	CheXpert Plus	bf16-equal	5/6	0.19	0.54	1.8	5.0
MedGemma 4B $\rightarrow$ MedGemma 27B	COCO	bf16-equal	3/6	0.05	0.35	0.6	4.6

$O_{\text{Pneumothorax}} = -0.71$ on CheXpert); in 27B, no direction moves the same three answers (-0.04 , -0.08 , -0.01). In 4B, the clean answer is yes on 94–97% of chest rows, so many cells sit at ceiling (clean answer above 0.99 or below 0.01 on more than 90% of rows) and $P(\text{yes})$ has little room to move. The logit margin $\ell_{\text{yes}} - \ell_{\text{no}}$ does not saturate, and ownership recomputed with margin changes in place of probability changes, O_q^m , runs from -1.1 to -19.8 with every interval below zero: every own write lowers its answer's margin by more than any competing write raises it. MedGemma 4B and 27B differ in the same way (Pneumothorax -0.60 versus -0.06 and Edema $+0.10$ versus -0.28 on CheXpert). The reader effect spans the whole matrix, including cells whose verdict stays the same. In the crossed swap of the Gemma 3 4B and MedGemma 4B towers and readers (Appendix A.10), changing the reader at a fixed tower changes O_q on NIH by a mean absolute 0.184, against 0.163 for changing the tower; the same appendix replays identical stored tensors to every reader of a shared tower.

Size and family. Ownership on the chest sets is concentrated among a few readers and varies by concept and dataset (Figure 7; Table 2). Qwen2.5-VL owns no concepts at 3B or 7B. At 32B, it owns Cardiomegaly; at 72B, Atelectasis ($O_{\text{Atelectasis}} = 0.54$), Cardiomegaly, and Mass. Effusion has negative ownership at every size. Lingshu, the medical Qwen2.5-VL derivative, owns three CheXpert concepts at 7B and five at 32B, the highest clinical ownership in the grid. LLaVA-1.5 reads the chest concepts but answers none above chance and owns none.

Medical and chest-specialised checkpoints. We divide the grid into three groups: checkpoints whose towers are trained on chest radiographs (CheXagent-2-3B, CheXagent-8B, LLaVA-Rad-7B), medically tuned checkpoints on general towers, and general checkpoints; HuatuoGPT-Vision-7B is tuned on medical data over a stock CLIP tower. The chest-trained towers' median probe AUROC across 18 NIH cells is 0.855, against 0.773 for HuatuoGPT-Vision's stock CLIP tower and 0.774

Table 15: **Alternative directions and competitor families.** Cells owned under the paper’s rule within each family, and the median O_q , per dataset. Each panel repeats the reference write on its own blocks.

Direction	Lift	NIH ChestX-ray14		CheXpert Plus		COCO	
		owned	med. O_q	owned	med. O_q	owned	med. O_q
<i>Direction construction (20 blocks per dataset)</i>							
logistic normal	coefficient	7/120	-0.03	19/120	-0.01	101/120	+0.18
difference of means	coefficient	16/120	-0.02	18/120	-0.01	62/120	+0.02
Haufe pattern	coefficient	17/120	-0.02	17/120	-0.01	53/120	+0.01
orthogonalised normal	coefficient	8/120	-0.02	14/120	-0.01	98/120	+0.13
residualised normal	coefficient	8/120	-0.02	13/120	-0.01	102/120	+0.20
owned under at least one		25/120	–	41/120	–	108/120	–
<i>Displacement lift (19 blocks per dataset)</i>							
logistic normal	coefficient	7/114	-0.03	19/114	-0.01	100/114	+0.19
difference of means	displacement	21/114	-0.03	19/114	-0.00	58/114	+0.02
Haufe pattern	displacement	17/114	-0.02	17/114	-0.00	49/114	+0.01
<i>Token weights of the logistic normal (4 NIH, 2 CheXpert, 2 COCO blocks)</i>							
uniform	coefficient	3/24	-0.08	5/12	+0.00	12/12	+0.60
softmax of the probe score	coefficient	0/24	-0.01	0/12	-0.00	12/12	+0.16
top quarter of tokens	coefficient	2/24	-0.12	3/12	-0.05	12/12	+0.57
<i>Competitor family of the logistic normal (5 NIH, 4 CheXpert blocks)</i>							
six clinical directions	coefficient	3/30	-0.07	8/24	-0.07	–	–
with every extra dataset label	coefficient	3/30	-0.07	5/24	-0.10	–	–

across the 132 remaining NIH cells. Across their 36 chest cells, they read 29 and answer 32. On NIH, the three groups own 2 of 18, 4 of 36, and 7 of 96 cells. The natural-image control of Section 5.3 rests on checkpoints that answer natural-image questions. The general group answers 80 of 96 COCO cells at a median clean-answer AUROC of 0.987 and owns 84; the medically tuned group answers 30 of 36 at 0.962 and owns 27. The chest-trained towers answer 6 of 18 COCO cells at 0.538 and own 2, so their COCO rows give the grade little to assess.

Non-clinical attributes. All 114 of 114 attribute cells are readable. Among answerable cells, attributes are owned in 11 of 100 and findings in 22 of 141. Pairing each attribute cell with clinical cells from its own block within 0.05 clean-answer AUROC yields 188 pairs, with attributes owned in 14 and findings in 17. On NIH, ownership is 7/57 for attributes and 6/114 for findings.

Answer direction. The answer direction’s write beats the strongest logistic competitor in 30 chest cells. Under the five held-out templates, a_q remains owned in 89 of 90 COCO (cell, template) pairs, and over all pairs it is owned in 87 of 180 chest and 89 of 90 COCO pairs. In Qwen2.5-VL-7B, it is owned for 3 of the six NIH concepts (for Effusion, its ownership against the other five answer directions is +0.29, with an own write of 0.81) and all 6 CheXpert concepts; in Lingshu 32B, for 3 NIH and 5 CheXpert concepts; in Gemma 3 12B, for 2 NIH and 0 CheXpert concepts. In Qwen2.5-VL-7B on NIH, the label AUROC is 0.60 for the answer direction and 0.73 for the probe, and the clean-answer AUROC is 0.93 and 0.66.

Cosine and projection depend on the inner product chosen for the representation (Park et al., 2024), so each cosine names its metric. The median raw model-space cosine between the answer and label directions spans 0.06–0.07 across the two chest blocks of Qwen2.5-VL-7B and 0.30–0.33 across those of Lingshu 32B, compared with 0.54–0.64 across the COCO blocks. For two coefficient vectors $\mathbf{u}, \mathbf{v}$ in the projected, train-scaled feature space, the whitened cosine $\cos_{\Sigma}(\mathbf{u}, \mathbf{v}) = \mathbf{u}^T \Sigma \mathbf{v} / \sqrt{\mathbf{u}^T \Sigma \mathbf{u} \mathbf{v}^T \Sigma \mathbf{v}}$, with Σ the covariance of the training features, is the correlation, over training images, between the scores that $\mathbf{u}$ and $\mathbf{v}$ assign to the train-scaled features. Between the answer direction’s and the probe’s coefficients, its median over each dataset’s cells is 0.36–0.43 on the two chest sets and 0.81 on COCO. The metrics differ most on the CheXpert block of Lingshu 32B, where the median over its six concepts is 0.33 raw and 0.87 whitened. The column-selectivity index

Table 16: **The answer direction.** Top, primary template: concepts owned by the label direction $\hat{w}_q$ and the answer direction a_q (kept, rescued, lost), and medians over the six questions. Bottom: a_q under each held-out template, with owned concepts, in parentheses those owned under IY, and kept (concept, template) pairs.

Checkpoint	Dataset	owned concepts					$a_q > \text{comp.}$	median			
		$\hat{w}_q$	a_q	kept	rescued	lost		$W_{q,q}^a$	cos	$\cos_\Sigma$	R^2
Qwen2.5-VL-7B	NIH ChestX-ray14	0/6	3/6	0	3	0	6/6	0.61	0.07	0.35	0.54
Qwen2.5-VL-7B	CheXpert Plus	0/6	6/6	0	6	0	6/6	0.55	0.06	0.36	0.48
Qwen2.5-VL-7B	COCO	6/6	6/6	6	0	0	6/6	0.78	0.64	0.84	0.58
Gemma 3 12B	NIH ChestX-ray14	1/6	2/6	1	1	0	5/6	0.47	0.18	0.33	0.17
Gemma 3 12B	CheXpert Plus	1/6	0/6	0	0	1	1/6	0.37	0.17	0.10	0.19
Gemma 3 12B	COCO	6/6	6/6	6	0	0	6/6	0.61	0.59	0.79	0.37
Lingshu 32B	NIH ChestX-ray14	2/6	3/6	2	1	0	6/6	0.24	0.30	0.60	0.77
Lingshu 32B	CheXpert Plus	5/6	5/6	5	0	0	6/6	0.22	0.33	0.87	0.80
Lingshu 32B	COCO	6/6	6/6	6	0	0	6/6	0.36	0.54	0.73	0.55
NIH ChestX-ray14	3 blocks	3/18	8/18	3	5	0	17/18	0.47	0.18	0.43	0.54
CheXpert Plus	3 blocks	6/18	11/18	5	6	1	13/18	0.35	0.16	0.36	0.48
<i>chest</i>	6 blocks	9/36	19/36	8	11	1	30/36	0.41	0.18	0.39	0.50
COCO	3 blocks	18/18	18/18	18	0	0	18/18	0.61	0.61	0.81	0.54

Checkpoint	Dataset	held-out template: owned (IY-owned kept)					kept pairs
		WY	IA	IB	WA	WB	
Qwen2.5-VL-7B	NIH ChestX-ray14	4 (3)	3 (3)	4 (3)	4 (3)	4 (3)	15/15
Qwen2.5-VL-7B	CheXpert Plus	4 (4)	6 (6)	5 (5)	5 (5)	5 (5)	25/30
Qwen2.5-VL-7B	COCO	6 (6)	6 (6)	6 (6)	6 (6)	6 (6)	30/30
Gemma 3 12B	NIH ChestX-ray14	2 (1)	1 (1)	0 (0)	1 (0)	1 (0)	2/10
Gemma 3 12B	CheXpert Plus	0 (0)	1 (0)	1 (0)	1 (0)	1 (0)	0/0
Gemma 3 12B	COCO	5 (5)	6 (6)	6 (6)	6 (6)	6 (6)	29/30
Lingshu 32B	NIH ChestX-ray14	3 (3)	3 (3)	3 (3)	3 (3)	3 (3)	15/15
Lingshu 32B	CheXpert Plus	5 (5)	3 (3)	3 (3)	4 (4)	4 (4)	19/25
Lingshu 32B	COCO	6 (6)	6 (6)	6 (6)	6 (6)	6 (6)	30/30
<i>chest</i>	6 blocks	18 (16)	17 (16)	16 (14)	18 (15)	18 (15)	76/95
COCO	3 blocks	17 (17)	18 (18)	18 (18)	18 (18)	18 (18)	89/90

$\kappa_d = W_{d,d} / \sum_q |W_{q,d}|$ divides a direction’s signed effect on its own question by its summed absolute effect on all six questions, so $\kappa_d = 1$ when a write moves only its own answer. Across the 36 chest cells, its median is 0.24 for answer directions and 0.19 for label directions in the same blocks, and 0.78 and 0.73 over the 18 COCO cells: on the chest sets, both answer and label directions move the other five answers along with their own. Over all cells, κ_d has median 0.12 on NIH and 0.12 on CheXpert, versus 0.57 on COCO (Table 19).

Semantic endpoints. The endpoint cells come from 3 scored blocks. The negated question moves opposite to the affirmative one in 16 of the 18 concept–block pairs. The forced choice is owned in both presentation orders in 3 of 18 pairs, with a mean absolute gap of 0.026–0.111 between orders. The report continuation is owned in 0 of 18 cells.

Labels and alternative constructions. On CheXpert, restricting evaluation to rows with a known label preserves the sign of O_q in 143 of 150 cells; NIH and COCO have every test label known, so their known-row numbers reproduce the all-row numbers. Refitting the six directions on the radiologist labels themselves leaves them a median raw model-space cosine of 0.12 and a median whitened cosine of 0.56 from the report-label directions, across the 6 CheXpert blocks with the refit. The expert-label refit reads the radiologist labels with a median cross-fitted AUROC of 0.69, versus 0.79 for the report-label direction on the same films, and a report-label direction fitted on as many rows as

Table 17: **Non-clinical attributes of the same radiographs.** Top: owned attribute and clinical cells in the nine-direction family per checkpoint and dataset, naming the owned attributes. Bottom: medians over blocks per attribute of probe AUROC, selectivity, clean-answer AUROC, and cosine to the clinical normals.

Checkpoint	Dataset	attributes owned	findings owned	view AP	sex F	age ≥ 60
Qwen2.5-VL-3B	NIH ChestX-ray14	0/3	0/6	–	–	–
Qwen2.5-VL-3B	CheXpert Plus	0/3	0/6	–	–	–
Qwen2.5-VL-7B	NIH ChestX-ray14	0/3	0/6	–	–	–
Qwen2.5-VL-7B	CheXpert Plus	0/3	0/6	–	–	–
Qwen2.5-VL-32B	NIH ChestX-ray14	0/3	0/6	–	–	–
Qwen2.5-VL-32B	CheXpert Plus	0/3	0/6	–	–	–
Qwen3-VL-4B	NIH ChestX-ray14	1/3	0/6	–	yes	–
Qwen3-VL-4B	CheXpert Plus	1/3	2/6	–	yes	–
Qwen3-VL-8B	NIH ChestX-ray14	0/3	0/6	–	–	–
Qwen3-VL-8B	CheXpert Plus	1/3	0/6	–	–	yes
Qwen3-VL-32B	NIH ChestX-ray14	0/3	1/6	–	–	–
Qwen3-VL-32B	CheXpert Plus	1/3	1/6	–	–	yes
InternVL3.5-8B	NIH ChestX-ray14	0/3	0/6	–	–	–
InternVL3.5-8B	CheXpert Plus	0/3	1/6	–	–	–
InternVL3.5-14B	NIH ChestX-ray14	0/3	1/6	–	–	–
InternVL3.5-14B	CheXpert Plus	0/3	1/6	–	–	–
InternVL3.5-38B	NIH ChestX-ray14	0/3	0/6	–	–	–
InternVL3.5-38B	CheXpert Plus	0/3	2/6	–	–	–
Gemma 3 4B	NIH ChestX-ray14	0/3	0/6	–	–	–
Gemma 3 4B	CheXpert Plus	0/3	0/6	–	–	–
Gemma 3 12B	NIH ChestX-ray14	0/3	1/6	–	–	–
Gemma 3 12B	CheXpert Plus	0/3	1/6	–	–	–
Gemma 3 27B	NIH ChestX-ray14	0/3	0/6	–	–	–
Gemma 3 27B	CheXpert Plus	0/3	0/6	–	–	–
MedGemma 4B	NIH ChestX-ray14	0/3	0/6	–	–	–
MedGemma 4B	CheXpert Plus	1/3	1/6	–	–	yes
MedGemma 27B	NIH ChestX-ray14	1/3	0/6	–	–	yes
MedGemma 27B	CheXpert Plus	0/3	0/6	–	–	–
Lingshu 7B	NIH ChestX-ray14	2/3	1/6	–	yes	yes
Lingshu 7B	CheXpert Plus	0/3	3/6	–	–	–
Lingshu 32B	NIH ChestX-ray14	2/3	2/6	–	yes	yes
Lingshu 32B	CheXpert Plus	1/3	4/6	–	–	yes
LLaVA-1.5-7B	NIH ChestX-ray14	0/3	0/6	–	–	–
LLaVA-1.5-7B	CheXpert Plus	0/3	0/6	–	–	–
LLaVA-1.5-13B	NIH ChestX-ray14	1/3	0/6	yes	–	–
LLaVA-1.5-13B	CheXpert Plus	0/3	0/6	–	–	–
LLaVA-Med 1.5 7B	NIH ChestX-ray14	0/3	0/6	–	–	–
LLaVA-Med 1.5 7B	CheXpert Plus	0/3	1/6	–	–	–
NIH ChestX-ray14	19 blocks	7/57	6/114	1/19	3/19	3/19
CheXpert Plus	19 blocks	5/57	17/114	0/19	1/19	4/19
chest	38 blocks	12/114	23/228	1/38	4/38	7/38
<i>Attribute probes and directions: medians over the blocks of each dataset</i>						
Attribute	Dataset	probe AUROC	selectivity	answer AUROC	med. cos	owned
AP (portable) projection	NIH ChestX-ray14	0.999	0.32	0.76	0.03	1/19
AP (portable) projection	CheXpert Plus	0.998	0.31	0.73	0.05	0/19
AP (portable) projection	chest	0.999	0.31	0.74	0.04	1/38
female sex	NIH ChestX-ray14	0.984	0.30	0.73	0.05	3/19
female sex	CheXpert Plus	0.957	0.27	0.65	0.06	1/19
female sex	chest	0.971	0.28	0.71	0.05	4/38
age ≥ 60	NIH ChestX-ray14	0.896	0.21	0.62	0.05	3/19
age ≥ 60	CheXpert Plus	0.898	0.21	0.64	0.06	4/19
age ≥ 60	chest	0.896	0.21	0.63	0.06	7/38
<i>The same comparison read three ways, pooled over the 38 chest blocks</i>						
Comparison	unit	attribute cells	owned	finding cells	owned	difference
every cell	cell	114	12	228	23	0.004
answerable cells only	cell	100	11	141	22	-0.046
answerability-matched pairs	pair	188	14	188	17	-0.016

Table 18: **Attributes and findings by clean-answer AUROC.** The nine-direction family’s cells binned by the clean answer’s AUROC against the question’s own label; each entry is owned cells of cells in the bin.

Cells	< 0.50	0.50–0.70	0.70–0.80	0.80–0.90	≥ 0.90	all cells
attributes, NIH ChestX-ray14	1/4	1/26	3/14	2/6	0/7	7/57
attributes, CheXpert Plus	0/5	3/29	0/12	2/5	0/6	5/57
<i>attributes, chest</i>	1/9	4/55	3/26	4/11	0/13	12/114
findings, NIH ChestX-ray14	0/6	2/63	2/32	2/13	0/0	6/114
findings, CheXpert Plus	0/13	4/47	2/9	5/21	6/24	17/114
<i>findings, chest</i>	0/19	6/110	4/41	7/34	6/24	23/228

Table 19: **The answer direction and column selectivity.** Top: per block, medians over the six questions of the AUROCs, the Pearson correlation, and the raw and whitened cosines between a_q and $\hat{w}_q$. Bottom: column selectivity $\kappa_d = W_{d,d}/\sum_q |W_{q,d}|$ per dataset, and ownership on the rows whose label for q is known.

Checkpoint	Dataset	AUROC, label		AUROC, clean answer		Pearson	cosine	
		a_q	$\hat{w}_q$	a_q	$\hat{w}_q$		raw	Σ
Qwen2.5-VL-7B	NIH ChestX-ray14	0.60	0.73	0.93	0.66	0.36	0.07	0.35
Qwen2.5-VL-7B	CheXpert Plus	0.67	0.86	0.85	0.66	0.34	0.06	0.36
Qwen2.5-VL-7B	COCO	0.97	0.97	0.98	0.98	0.84	0.64	0.84
Lingshu 32B	NIH ChestX-ray14	0.75	0.76	0.95	0.79	0.62	0.30	0.60
Lingshu 32B	CheXpert Plus	0.85	0.86	0.95	0.89	0.88	0.33	0.87
Lingshu 32B	COCO	0.94	0.97	0.93	0.92	0.75	0.54	0.73
Gemma 3 12B	NIH ChestX-ray14	0.59	0.65	0.72	0.61	0.31	0.18	0.33
Gemma 3 12B	CheXpert Plus	0.61	0.78	0.73	0.56	0.13	0.17	0.10
Gemma 3 12B	COCO	0.93	0.96	0.80	0.76	0.81	0.59	0.79

Alignment against ownership over the 54 answer-direction cells: Spearman correlation between $\cos_\Sigma(a_q, \hat{w}_q)$ and the label direction’s O_q $\rho = 0.75$ ($p = 8 \times 10^{-11}$); with the label direction’s owned indicator $\rho = 0.74$ ($p = 1 \times 10^{-10}$); median $\cos_\Sigma$ 0.80 in owned against 0.34 in not-owned cells; within the chest sets alone $\rho = 0.59$ ($p = 2 \times 10^{-4}$, 36 cells).

Column selectivity and known-label ownership, per dataset

Dataset	blocks	cells	med. κ_d	$\kappa_d \geq 0.5$	known rows	sign kept	owned kept
NIH ChestX-ray14	25	150	0.12	2/150	600	150/150	13/13
CheXpert Plus	25	150	0.12	2/150	165	143/150	23/25
COCO	25	150	0.57	92/150	600	150/150	113/113

the refit reaches 0.70: the estimation sample moves the result by +0.09 and the label source by −0.01. The difference-of-means and pattern directions sit at a median absolute raw model-space cosine of 0.04 to 0.28 from the logistic normal, and each is owned in more NIH cells and fewer COCO cells than the logistic normal. Lifted as displacements, they are owned in 40 and 34 of 228 chest cells and in 58 and 49 of 114 COCO cells, respectively (Table 15). Every construction preserves the chest-to-COCO gap: at least one construction is owned in 25 of 120 cells on NIH and 41 of 120 on CheXpert, compared with 108 of 120 on COCO. The alternative constructions are scored across 60 blocks, the extended competitor family across 9 blocks, and the token-weighted writes across 6 chest blocks; on COCO, the softmax and top-quarter writes are owned in 12 and 12 of 12 cells.

A.10 Crossing the factors, replaying the tensors, and changing the endpoint

Crossed tower and reader. The swap replaces the host checkpoint’s vision tower with the donor checkpoint’s tower, leaving the host’s connector and language model untouched. For every swap, the module records a receipt listing the replaced tensors of the tower root, their parameter count, a digest of the donor tower, and tensor-by-tensor comparisons of the loaded tower against the donor and of everything outside the tower against the host’s own weights. Of the host tower’s 438 tensors, 437 carry weights; the module replaces each and verifies bitwise equality with the donor’s (417 million parameters). The module rebuilds the remaining tensor, a position buffer. None of the 451 tensors outside the tower changes. Write directions are fitted in the representation each tower produces, so a swapped arm writes the donor’s directions. The two factors therefore carry different loads: at

Table 20: **Radiologist labels.** Top: the complete grade per CheXpert block on 200 radiologist-labelled films, against the same grade on the 600 labeler-labelled test rows. Bottom: the six directions refitted on the radiologist labels.

Checkpoint	owned (test)	owned (valid)	verdict agrees	owned grade agrees	median $ \Delta O_q $
Qwen2.5-VL-3B	0	0	5/6	6/6	0.005
Qwen2.5-VL-7B	0	0	6/6	6/6	0.005
Qwen2.5-VL-32B	0	0	6/6	6/6	0.004
Qwen3-VL-4B	2	2	4/6	6/6	0.009
Qwen3-VL-8B	0	0	5/6	6/6	0.003
Qwen3-VL-32B	1	1	6/6	6/6	0.004
InternVL3.5-8B	1	0	5/6	5/6	0.001
InternVL3.5-14B	1	1	3/6	6/6	0.002
InternVL3.5-38B	2	2	6/6	6/6	0.001
Gemma 3 4B	0	0	4/6	6/6	0.007
Gemma 3 12B	1	1	6/6	6/6	0.016
Gemma 3 27B	0	0	3/6	6/6	0.002
MedGemma 4B	2	1	5/6	5/6	0.025
MedGemma 27B	0	0	5/6	6/6	0.010
Lingshu 7B	3	3	6/6	6/6	0.007
Lingshu 32B	5	5	6/6	6/6	0.005
LLaVA-1.5-7B	0	0	5/6	6/6	0.003
LLaVA-1.5-13B	0	0	5/6	6/6	0.000
LLaVA-Med 1.5 7B	1	1	6/6	6/6	0.000
<i>all</i> (19 blocks)	19	17	97/114	112/114	0.003

The label source of a refitted direction against the number of rows it was fitted on. The expert-label refit is estimated on 200 radiologist-labelled films and the shipped direction on about 20,000 report-labelled rows, so comparing only those two confounds the two. Every arm keeps the block's own projection, its stored train-only scaler, the protocol's probe settings and the same 5-fold cross-fitting over patients, and is scored on the same films by a direction not fitted on them. The two training subsample arms redraw the fitting rows 20 times and report the median; the second draws enough rows to leave the concept with as many labelled rows as the expert refit has. Cosines are to the shipped direction. Medians over the 36 cells of 6 CheXpert blocks.

Labels	Fitted on	cross-fitted AUROC against		cosine to shipped	
		rows	radiologist	report	raw
radiologist	200 valid films	160	0.69	0.71	0.12
report-derived	the same 200 valid films	44	0.67	0.71	0.12
report-derived	200 training rows	42	0.66	0.70	0.12
report-derived	200 labelled training rows	160	0.70	0.74	0.22
report-derived	every training row	5332	0.79	0.86	1.00
<i>What each contrast isolates, in AUROC against the radiologist labels</i>					
label source, both arms fitting the same number of rows					-0.01
sample size, both arms on report-derived labels					0.09
the expert refit against the shipped direction, which mixes the two					-0.11

a fixed tower the swap changes the connector and the language model and nothing else, while replacing the tower replaces the representation and the six directions fitted on it together. The tower factor is the joint effect of both; the reader factor is the effect of neither. Each of the 4 combinations of tower and reader is graded on the same 200 test rows of the block, at the primary dose and template, against the 119-direction random family and the sham of the tower in use, with the 6×5 max- T verdict (Table 10). The two native combinations give the two checkpoints' own grades on that cohort. Across the 36 native cells, the ownership contrast changes by at most 0.060 relative to the grade on the block's whole test cohort. On the smaller cohort, 3 of the 13 cells owned in that grade become unresolved. The crossover contrasts hold one factor fixed and change the other, and report three quantities of the same six questions: the own write $W_{q,q}$, the ownership contrast O_q , and the margin of the own write over the strongest competitor in the whole reference family, $M_q = W_{q,q} - \max(\max_{d \neq q} W_{q,d}, \text{random } p_{95}, |\text{sham}|)$, each as the mean over the six questions of the absolute paired change, with a percentile interval over the same 2,000 draws of the rows and, in brackets, the number of questions of six whose change is simultaneously nonzero. The last two rows

of each block average each factor over its two levels. The three come from one pass over the four arms’ packaged outcomes, so a row’s columns are the same arms read three ways. The four arms of a dataset are the same four arms whichever of the pair’s blocks indexes them, and the two host blocks give the same crossover: the middle panel counts each dataset’s four combinations once, and an arm graded in both host blocks agrees in all 10 cases where it appears twice. A dash marks an arm that is not graded in that host block. Table 21 reports every combination and contrast.

What the crossing owns. Gemma 3 4B and MedGemma 4B give 4 combinations of tower and reader on each dataset, and written directions travel with the tower that produced them. On NIH ChestX-ray14 and CheXpert Plus, no combination owns a finding: 0 of 48 cells across the 8 combinations. On COCO, the same crossing owns 14 of 24 cells. Each native combination owns 5 of the six objects, the Gemma tower behind the MedGemma reader owns 4, and the MedGemma tower behind the Gemma 3 reader owns 0. The crossover holds one factor fixed and changes the other on three quantities of the same six questions. On NIH, the mean absolute reader effect is 0.186 [0.170, 0.207] on the own write and 0.184 [0.169, 0.203] on the ownership contrast, against a tower effect of 0.162 [0.140, 0.182] and 0.163 [0.149, 0.185]. On CheXpert the reader effect is 0.254 on the own write and 0.259 on the ownership contrast, against 0.226 and 0.260 for the tower. On the family margin the two chest sets give 0.356 and 0.365 for the reader against 0.316 and 0.444 for the tower. On COCO the tower effect is the larger of the two on all three quantities: 0.248, 0.416, and 0.353 against 0.137, 0.327, and 0.288.

How well the crossed arms answer. The hybrid arms answer worse than the native arms. On the labelled rows the crossing grades, with no write applied, the median clean-answer AUROC over a block’s six concepts falls from 0.930 in the native Gemma 3 4B arm on COCO to 0.488 in the MedGemma tower behind that reader, from 0.840 in the native MedGemma 4B arm on NIH ChestX-ray14 to 0.549 in the Gemma 3 tower behind that reader, and from 0.889 to 0.636 on CheXpert Plus. The remaining three arms give 0.506 against 0.425 and 0.560 against 0.440 in the Gemma 3 4B blocks, and 0.951 against 0.865 in the MedGemma 4B block on COCO. All 6 of the 6 hybrid arms fall below the native arm of their own block, by 0.081 to 0.442.

Identical-tensor replay. Checkpoints within one family share a vision tower, but two readers’ inputs agree only up to bf16 rounding. The replay module removes this rounding difference. It stores the consumed block’s token tensors once from the group’s source checkpoint and feeds them to every reader in the group, replacing each reader’s own tower output so the two readers consume the same bits. Before replacement, it measures drift as the largest absolute difference between the reader’s own tower output and the stored tensor on the same rows. A reader replaying the tensors of its own block controls for the machinery: its replayed grade is its own grade computed through the replacement path. The module also recomputes each pair’s between-reader difference twice: once with each reader running its own tower and once with both reading the stored tensors. The two computations differ only in their input. Across the 3 replays in which a reader reads its own block’s tensors, no verdict and no ownership grade changes; 2 reproduce the block’s own grade exactly and the third differs by at most 6.2×10^{-3} in a write magnitude. In 7 of the 8 replays the reader’s own tower output already matches the stored tensor bit for bit. The exception is LLaVA-Med, whose fine-tuned tower departs from it by up to 112.16, against a mean absolute tensor entry of 0.62–29.12. Across the 8 replays, the change is distinguishable from zero in 5 of 48 concept cells, the largest change in a write magnitude is 1.5×10^{-2} , and 0 verdicts and 0 ownership grades change. Across the 7 reader pairs, the mean absolute difference between the readers’ own-question write magnitudes is 0.016–0.348 when each reader runs its own tower, and feeding both readers the stored tensors changes that difference by at most 0.0052, or 1.9% of its size.

Semantic endpoints. The four endpoints change what is read after the write, not what is written. The negated question asks whether there is no evidence of the finding. A direction that carries the concept raises the affirmative answer and lowers the negated one, so the two raw effects must have

Table 21: **Crossing the tower and the reader.** Each column of the top two panels is tower/reader, with G for Gemma 3 4B and M for MedGemma 4B. The bottom panel changes one factor at a time and gives the mean absolute paired change over the six questions.

Host checkpoint	Dataset	native		swapped		concepts	crossover
		G/G	M/M	M/G	G/M		
Gemma 3 4B	NIH ChestX-ray14	0	0	0	0	6	complete
Gemma 3 4B	CheXpert Plus	0	0	0	0	6	complete
Gemma 3 4B	COCO	5	5	0	4	6	complete
MedGemma 4B	NIH ChestX-ray14	0	0	0	0	6	complete
MedGemma 4B	CheXpert Plus	0	0	–	0	6	–
MedGemma 4B	COCO	5	5	–	4	6	–
<i>The four combinations of each dataset, shared arms counted once</i>							
Combinations	Dataset	G/G	M/M	M/G	G/M	concepts	owned of cells
all four	NIH ChestX-ray14	0	0	0	0	6	0/24
all four	CheXpert Plus	0	0	0	0	6	0/24
all four	COCO	5	5	0	4	6	14/24
<i>Crossover: the effect of replacing one factor with the other held fixed</i>							
Block	Changed, held	own write $W_{q,q}$		ownership O_q		family margin M_q	
Gemma 3 4B, NIH	reader, own tower	0.133	[0.106, 0.169] (1)	0.158	[0.136, 0.187] (4)	0.425	[0.393, 0.458] (5)
	reader, partner tower	0.240	[0.221, 0.259] (5)	0.210	[0.192, 0.233] (4)	0.286	[0.268, 0.309] (6)
	tower, own reader	0.164	[0.140, 0.189] (4)	0.061	[0.046, 0.080] (1)	0.137	[0.120, 0.164] (4)
	tower, partner reader	0.160	[0.125, 0.193] (6)	0.265	[0.239, 0.299] (5)	0.496	[0.462, 0.525] (6)
	mean reader effect	0.186	[0.170, 0.207]	0.184	[0.169, 0.203]	0.356	[0.339, 0.376]
	mean tower effect	0.162	[0.140, 0.182]	0.163	[0.149, 0.185]	0.316	[0.299, 0.338]
Gemma 3 4B, CheXpert	reader, own tower	0.382	[0.353, 0.412] (5)	0.303	[0.273, 0.332] (6)	0.490	[0.465, 0.515] (5)
	reader, partner tower	0.125	[0.102, 0.151] (4)	0.214	[0.194, 0.240] (5)	0.240	[0.214, 0.263] (6)
	tower, own reader	0.269	[0.245, 0.293] (5)	0.223	[0.205, 0.241] (6)	0.372	[0.352, 0.390] (5)
	tower, partner reader	0.183	[0.153, 0.217] (4)	0.297	[0.267, 0.332] (5)	0.517	[0.485, 0.553] (6)
	mean reader effect	0.254	[0.235, 0.274]	0.259	[0.240, 0.280]	0.365	[0.347, 0.383]
	mean tower effect	0.226	[0.208, 0.246]	0.260	[0.243, 0.279]	0.444	[0.425, 0.466]
Gemma 3 4B, COCO	reader, own tower	0.072	[0.049, 0.104] (2)	0.229	[0.187, 0.271] (5)	0.143	[0.114, 0.185] (3)
	reader, partner tower	0.202	[0.170, 0.239] (3)	0.425	[0.390, 0.459] (6)	0.432	[0.398, 0.464] (6)
	tower, own reader	0.295	[0.261, 0.337] (5)	0.497	[0.453, 0.540] (6)	0.408	[0.368, 0.450] (6)
	tower, partner reader	0.200	[0.175, 0.227] (4)	0.335	[0.307, 0.364] (5)	0.298	[0.273, 0.326] (5)
	mean reader effect	0.137	[0.119, 0.161]	0.327	[0.298, 0.354]	0.288	[0.265, 0.315]
	mean tower effect	0.248	[0.228, 0.271]	0.416	[0.392, 0.441]	0.353	[0.331, 0.379]
MedGemma 4B, NIH	reader, own tower	0.240	[0.221, 0.259] (5)	0.210	[0.192, 0.233] (4)	0.286	[0.268, 0.309] (6)
	reader, partner tower	0.133	[0.106, 0.169] (1)	0.158	[0.136, 0.187] (4)	0.425	[0.393, 0.458] (5)
	tower, own reader	0.160	[0.125, 0.193] (6)	0.265	[0.239, 0.299] (5)	0.496	[0.462, 0.525] (6)
	tower, partner reader	0.164	[0.140, 0.189] (4)	0.061	[0.046, 0.080] (1)	0.137	[0.120, 0.164] (4)
	mean reader effect	0.186	[0.170, 0.207]	0.184	[0.169, 0.203]	0.356	[0.339, 0.376]
	mean tower effect	0.162	[0.140, 0.182]	0.163	[0.149, 0.185]	0.316	[0.299, 0.338]

opposite signs. The forced choice asks which is present: the finding or its strongest competitor, taken from the block’s own primary-template write matrix. We ask it in both presentation orders, so a position preference appears as a gap between them. The report continuation requests the findings section and reads the log-probability of the finding word. Each endpoint carries its own clean baseline, a coordinate-permutation sham of each direction under test, and its own family of 18 random directions written at the same dose. An endpoint cell meets the steering reference when its own signed effect is positive, above every one of those 18 random writes, and above the absolute sham, and it is owned when it also holds the campaign’s simultaneous 6×5 max- T advantage. The random bar here is the maximum of a family of 18 rather than the 95th percentile of the campaign’s 119; everything else is the campaign’s rule (Table 10). The incremental-validity test regresses the endpoint effect on write magnitude, probe selectivity, and clean-answer AUROC, with endpoint dummies, then adds the

ownership contrast and its owned indicator. The penalty and the standardisation are fitted inside each training fold, so no held-out cell enters its own prediction. The predictors are per-concept block-level quantities, so the effective sample is the six concepts rather than the cells. The patient bootstrap preserves the dependence between cells that share rows. We cross-validate both models twice, leaving out one concept at a time inside a block and one block at a time across the 3 blocks. Held-out R^2 changes by +0.687 under the concept split and +0.070 under the block split. A null that permutes ownership across the concepts of a block over 2,000 draws puts both at $p = 0.26$ and $p = 0.26$, and no single split reaches $p < 0.05$. Table 22 reports every endpoint, test, and regression.

Table 22: **Four endpoints no direction was fitted against.** Both direction families of Section 5.6, written at the primary dose and read at the negated question, the forced choice in both orders, and a report continuation. Bottom: the ownership contrast added to a regression of the endpoint effect, cross-validated by concept and by block.

Checkpoint	Dataset	Endpoint	label direction		answer direction	
			owned	reference	owned	reference
Lingshu 32B	CheXpert Plus	negated question	2/6	2/6	3/6	4/6
Lingshu 32B	CheXpert Plus	forced choice, finding first	3/6	3/6	4/6	4/6
Lingshu 32B	CheXpert Plus	forced choice, competitor first	4/6	4/6	5/6	5/6
Lingshu 32B	CheXpert Plus	report continuation	0/6	0/6	0/6	0/6
Qwen2.5-VL-7B	NIH ChestX-ray14	negated question	0/6	1/6	3/6	6/6
Qwen2.5-VL-7B	NIH ChestX-ray14	forced choice, finding first	0/6	0/6	3/6	3/6
Qwen2.5-VL-7B	NIH ChestX-ray14	forced choice, competitor first	0/6	0/6	2/6	2/6
Qwen2.5-VL-7B	NIH ChestX-ray14	report continuation	0/6	0/6	0/6	1/6
Qwen2.5-VL-7B	CheXpert Plus	negated question	1/6	1/6	5/6	6/6
Qwen2.5-VL-7B	CheXpert Plus	forced choice, finding first	0/6	0/6	3/6	4/6
Qwen2.5-VL-7B	CheXpert Plus	forced choice, competitor first	0/6	0/6	2/6	2/6
Qwen2.5-VL-7B	CheXpert Plus	report continuation	0/6	0/6	0/6	0/6
<i>all</i>	<i>3 blocks</i>	<i>4 endpoints</i>	10/72	11/72	30/72	37/72
<i>The three endpoint-specific tests</i>						
Checkpoint	Dataset	opposite on the negation	order gap	forced choice owned	report owned	core owned
Lingshu 32B	CheXpert Plus	4/6	0.026	3/6	0/6	5/6
Qwen2.5-VL-7B	NIH ChestX-ray14	6/6	0.089	0/6	0/6	0/6
Qwen2.5-VL-7B	CheXpert Plus	6/6	0.111	0/6	0/6	0/6
<i>What ownership adds, held out: leaving out one concept at a time</i>						
Checkpoint	Dataset	Direction	held-out R^2	with ownership	ΔR^2	permutation p
Lingshu 32B	CheXpert Plus	label direction	-0.034	0.147	0.181	0.39
Lingshu 32B	CheXpert Plus	answer direction	0.069	-0.223	-0.291	0.70
Qwen2.5-VL-7B	CheXpert Plus	label direction	-0.139	-0.149	-0.010	0.55
Qwen2.5-VL-7B	CheXpert Plus	answer direction	-3.168	-0.112	3.057	0.18
Qwen2.5-VL-7B	NIH ChestX-ray14	label direction	-0.195	-0.147	0.048	0.31
Qwen2.5-VL-7B	NIH ChestX-ray14	answer direction	-1.310	-0.176	1.134	0.50
<i>pooled</i>	<i>6 tests</i>	<i>both</i>	—	—	0.687	0.26
<i>The same, leaving out one block at a time</i>						
<i>all 3</i>	<i>pooled</i>	label direction	-0.578	-0.500	0.078	0.30
<i>all 3</i>	<i>pooled</i>	answer direction	-0.461	-0.398	0.063	0.26
<i>pooled</i>	<i>2 tests</i>	<i>both</i>	—	—	0.070	0.26
<i>In-sample reference only, since adding a predictor cannot lower a training R^2</i>						
Checkpoint	Dataset	Direction	training R^2	with ownership	ΔR^2	
Lingshu 32B	CheXpert Plus	label direction	0.769	0.788	0.019	
Lingshu 32B	CheXpert Plus	answer direction	0.368	0.423	0.055	
Qwen2.5-VL-7B	NIH ChestX-ray14	label direction	0.241	0.246	0.004	
Qwen2.5-VL-7B	NIH ChestX-ray14	answer direction	0.267	0.392	0.125	
Qwen2.5-VL-7B	CheXpert Plus	label direction	0.175	0.250	0.074	
Qwen2.5-VL-7B	CheXpert Plus	answer direction	0.473	0.509	0.036	

Fine-grained objects and difficulty matching. The rule that picks the six additional COCO categories, and the two matching windows, are stated in full in Appendix F.6. The rule reads the label counts and the mean instance areas of the fixed cohorts and nothing else; the registered protocol file carries it under the date of amendment A15, and the module’s first outcome shard is later than that date, so the categories and the windows were fixed before any of them was scored and before any

ownership outcome of theirs existed. We fit, lift, write, and grade these categories exactly like the six easy objects, in the same blocks and on the same rows, as their own six-direction family, against a random family and a sham of their own (Table 10). We measure their clean-answer AUROC on the same calibration rows and by the same rule as every other cell, giving the matching variable the same meaning on both sides. The matched comparison pairs each chest cell with every natural-image cell whose matching variables lie within 0.05 of it. It reports the ownership rate on each side of the surviving pairs under two estimators. The *pooled* difference is the ownership rate among the matched chest cells minus the rate among the natural-image cells that served as a partner at least once. The *matched* difference is the mean over matched chest cells of that cell’s own ownership minus the ownership rate of its own partner set, so a chest cell with many partners and one with few count equally. Neither estimator can be obtained from the other. The *chest cells* and *partners* columns give the effective sample of both: matched chest cells of all chest cells, and distinct natural-image cells used as a partner of all of them. Both intervals are percentile intervals over 5,000 draws of a cluster bootstrap that resamples models, keeping each model’s chest and natural-image cells together. Chest cells without a partner leave the comparison. The stratified comparison answers the same question without dropping a cell. It bins every cell of both pools by its clean-answer AUROC at the edges 0.50, 0.70, 0.80 and 0.90, fixed in the released analysis code, and reports the ownership rate inside each bin with the same cluster bootstrap over models on the difference in the top bin. The last bin is closed on the right and every cell of each pool falls in exactly one bin. The 37 chest cells of that bin are CheXpert Plus cells; no NIH ChestX-ray14 cell reaches a clean-answer AUROC of 0.90. The same stratification of the nine-direction attribute family appears in Table 17. Table 23 reports the per-block counts, the three pools, each matching, and every bin.

A.11 Write structure across the grid

Figure 8 summarises the 6×6 write matrices across all completed checkpoints as the cell-wise median matrix for each dataset. Figure 3 reads the same matrices as flow: band width is proportional to $F_{q,d}$, the mean positive answer change across checkpoints, coloured bands reach the writer’s own question, and grey bands reach other questions. The write-flow share is the total width of own-question bands divided by the total width of all bands, $\sum_d F_{d,d} / \sum_{q,d} F_{q,d}$: 24% on NIH and 22% on CheXpert, compared with 79% on COCO. The column-selectivity index κ_d in Table 19 measures a different quantity. It retains the sign of a direction’s own effect. Its denominator includes negative changes. It gives every cell equal weight. We report its median over cells (0.12, 0.12, and 0.57). The largest sinks on the chest sets are the Cardiomegaly question on NIH and the Edema and Effusion questions on CheXpert. These questions receive writes from every direction.

A.12 Per-image examples

Figure 9 shows grades for individual test images, with $P(\text{yes})$ in parentheses on every answer line. The first four chest examples are deficit cells of Qwen2.5-VL-7B, the next two are owned cells (Lingshu 32B Edema, Qwen2.5-VL-72B Atelectasis), and the natural-image examples are owned cells of Qwen2.5-VL-7B, Lingshu 32B, and MedGemma 4B. For a chest example, we pool test rows where the finding is labelled present, the probe reads it (probability above 0.5), and the clean answer is No. For a natural-image example, we pool rows where the probe reads the label correctly and the clean answer is No. We split each pool by whether the concept’s own write or the strongest competing write raises the answer more. Among rows where either write raises the answer by at least 0.2, we show the row with the gap closest to the median. Each example’s last line reports how many pool rows share its ordering, giving the example its own base rate.

B Pilot study: registered protocol

The two-model study that motivated the evaluation registered the procedures below; Appendix C reports its results.

Table 23: **Small and fine-grained objects, and difficulty matching.** Top and middle: six fine-grained COCO categories graded like the six easy objects, against the easy objects of every COCO block and of the 6 blocks with fine-grained cells. Bottom: chest cells matched to natural-image cells on difficulty, and the same cells stratified on clean-answer AUROC.

Checkpoint	owned of six		median answer AUROC		median selectivity	
	fine	easy	fine	easy	fine	easy
Qwen2.5-VL-3B	6	6	0.947	0.971	0.024	0.037
Qwen3-VL-4B	5	6	0.984	0.990	0.036	0.033
InternVL3.5-8B	6	6	0.981	0.978	0.104	0.117
Gemma 3 4B	4	5	0.948	0.924	0.076	0.070
Lingshu 7B	6	6	0.950	0.968	0.025	0.023
LLaVA-1.5-7B	3	5	0.956	0.979	0.266	0.285

Paired within these blocks: fine-grained minus easy is +0, -1, +0, -1, -2, +0 cells of six; fine higher in 0, easy higher in 3, tied in 3; exact two-sided sign test $p = 0.25$

<i>The four pools of cells</i>						
Cells	blocks	owned	rate	median answer AUROC	median selectivity	readable
chest findings	50	38/300	0.127	0.695	0.137	178/300
easy objects, every COCO block	25	113/150	0.753	0.977	0.064	90/150
easy objects, these blocks	6	34/36	0.944	0.980	0.065	24/36
fine-grained objects, these blocks	6	30/36	0.833	0.963	0.058	23/36

<i>Chest cells matched to natural-image cells on difficulty, under both estimators</i>						
Matching	chest cells	partners	chest rate	natural rate	difference	95% interval
probe selectivity alone	300/300	186/186	0.127	0.769	-0.642	[-0.763, -0.512]
pooled matched					-0.539	[-0.678, -0.359]
clean-answer AUROC alone	300/300	186/186	0.127	0.769	-0.642	[-0.749, -0.505]
pooled matched					-0.066	[-0.304, -0.004]
both variables	212/300	101/186	0.137	0.624	-0.487	[-0.653, -0.201]
pooled matched					-0.024	[-0.198, 0.076]
both variables, easy partners only	211/300	79/150	0.137	0.570	-0.432	[-0.625, -0.150]
pooled matched					-0.008	[-0.198, 0.085]

<i>Chest cells and natural-image cells stratified on clean-answer AUROC, owned of cells</i>						
Cells	< 0.50	0.50–0.70	0.70–0.80	0.80–0.90	≥ 0.90	all cells
NIH ChestX-ray14	0/7	6/80	3/43	4/19	0/1	13/150
CheXpert Plus	0/13	6/52	2/20	6/29	11/36	25/150
chest findings	0/20	12/132	5/63	10/48	11/37	38/300
easy objects, every COCO block	0/6	0/6	0/7	1/4	112/127	113/150
fine-grained objects, these blocks	0/0	0/0	0/0	1/1	29/35	30/36
natural-image objects	0/6	0/6	0/7	2/5	141/162	143/186

chest minus natural-image in the ≥ 0.90 bin: -0.573, 95% interval [-0.763, -0.388]

Data and split. We use NIH ChestX-ray14 (Wang et al., 2017), whose findings are mined from radiology reports. A deterministic hash of patient identity assigns rows to disjoint train, validation, and test partitions. The registered manifest contains 18,212 training images and 5,343 test images from 1,659 test patients. The three cells pair LLaVA-1.5-7B with Effusion and Edema, and Qwen2.5-VL-7B with Effusion. All reported uncertainty clusters by patient. Prompts appear in Appendix D.

Models and consumed loci. For LLaVA-1.5-7B (Liu et al., 2024) and Qwen2.5-VL-7B (Bai et al., 2025b), we use the final visual block consumed by the connector or merger. Writing α for the signed intervention dose, synthetic hook tests establish downstream-path membership, a bitwise $\alpha = 0$ no-op, and changed downstream representations and logits at nonzero dose. Probe and intervention therefore share one consumed representation.

Controlled decodability. For image i , let $\mathbf{h}_{it} \in \mathbb{R}^D$ denote the D -dimensional representation of visual token t at the consumed final block; mean pooling over tokens yields $\mathbf{h}_i$. A fixed seed-0 Gaussian matrix $R \in \mathbb{R}^{D \times 512}$ projects every model to a common readout dimension; featurewise scale s is fitted on training rows only. We then fit logistic regression with inverse regularisation strength $C = 1$, maximum 2,000 iterations, and seed 0. Each of 20 control seeds maps the recurring input type (AP view indicator, sex indicator, and age decade, giving 39 strata) to a random label. Let y be the real test labels and r the fitted probe’s continuous scores. For control seed $k \in \{0, \dots, 19\}$,

y_k^{ctrl} denotes the random labels and r_k^{ctrl} the corresponding control probe scores. Selectivity compares their AUROCs:

$$S = \text{AUROC}(y, r) - \frac{1}{20} \sum_{k=0}^{19} \text{AUROC}(y_k^{\text{ctrl}}, r_k^{\text{ctrl}}). \quad (5)$$

We report the real AUROC, control distribution, and S with 2,000 patient-cluster bootstrap resamples. A positive interval for S means that the target is more readable than same-capacity nuisance-type label maps. It does not yet imply downstream use. Paired contrasts reuse identical ordered test rows and bootstrap draws.

Probe-normal intervention. Let β_c be the projected-space logistic coefficient for the positive class of concept c . We lift it to the consumed D -dimensional representation and normalize it exactly as

$$\hat{\mathbf{w}}_c = \frac{\mathbf{g}_c}{\|\mathbf{g}_c\|_2}, \quad \mathbf{g}_c = R \text{diag}(\mathbf{s})^{-1} \beta_c, \quad (6)$$

with every entry of $\mathbf{s}$ floored at 10^{-8} . The positive-class coefficient fixes the direction’s clinical pole. At every visual token t , the registered intervention is

$$\mathbf{h}'_{it} = \mathbf{h}_{it} + \alpha \|\mathbf{h}_{it}\|_2 \hat{\mathbf{w}}_c. \quad (7)$$

Here the sign of α selects the positive or negative clinical pole, while $|\alpha|$ is the perturbation norm as a fraction of each token’s activation norm. At the first answer position, $P(\text{yes}) = \sigma(\ell_{\text{yes}} - \ell_{\text{no}})$, where each log probability is the maximum over case and leading-space variants that tokenize to one token; the endpoint is its paired mean change from the common unsteered baseline. Original cells use concept doses $\{-1, -0.5, -0.25, -0.1, 0, 0.1, 0.25, 0.5, 1\}$ on 200 held-out test images. Controls are evaluated at the six nonzero doses excluding ± 0.1 : 20 isotropic random unit directions, one sham formed by a fixed coordinate permutation of $\hat{\mathbf{w}}_c$ and renormalization, and fixed alternative clinical probe normals. Every control receives the same signed dose on the same rows. Dose monotonicity over $[-0.5, 0.5]$ is descriptive.

Original-cell decision rule. Let $\Delta_c(\alpha)$ be the target direction’s paired mean probability change, and let $\mathcal{A} = \{-1, -0.5, -0.25, 0.25, 0.5, 1\}$ be the six controlled doses. The primary effect is $\max_{\alpha \in \mathcal{A}} \text{sign}(\alpha) \Delta_c(\alpha)$, attained at dose α^* . A cell clears the generic intervention gate when this selected effect exceeds the 95th percentile of the 20 random directions’ mean probability changes at that same α^* , each multiplied by $\text{sign}(\alpha^*)$, and the maximum absolute sham effect over $\mathcal{A}$. The random directions are not separately maximized over doses. This asymmetric selection defines a registered reference gate, not a cross-dose-corrected 5% test. The conjunction is fixed across the three cells. The independent supplement below instead evaluates a dose locked before observing its patients.

Direction-specificity supplement. The initial experiment prespecified the random and sham comparisons; its clinical-direction analysis was descriptive. The Nodule response motivated a separately registered supplement at the locked Qwen dose, $\alpha = +0.25$. We select one row by label-blind hash for each of 400 ChestX-ray14 test patients absent from the original 200-image intervention cohort. The semantic family contains all five non-target normals from the pre-existing Qwen direction bundle (Atelectasis, Pneumothorax, Cardiomegaly, Mass, and Nodule), rather than directions chosen from the supplement outcome. Each uses the same training rows, projection, scaling, classifier, lifting rule, and normalization as Effusion. Together with a common baseline, 20 random directions, and the sham, the grid produces 28 configurations and 11,200 per-image outcomes; exact row selection appears in Appendix C.

Semantic ownership is a contrast on a direction–question response surface, not a property of a direction in isolation. Let $W_{q,d}$ denote the paired mean change in question q ’s yes probability after

intervening with direction d at the locked dose. A large $W_{q,d}$ establishes that d can write q 's answer; it does not establish that d owns that answer. Within a fixed clinical family, the ownership contrast is

$$O_q = W_{q,q} - \max_{d \neq q} W_{q,d}. \quad (8)$$

The registered supplement estimates the Effusion row of this surface, so the reported ownership contrast O_{Effusion} is Effusion minus its maximum alternative. The maximum is recomputed inside each of 5,000 patient-bootstrap draws. Thus $O_{\text{Effusion}} > 0$ means Effusion beats every member of the fixed clinical family, while $O_{\text{Effusion}} < 0$ means at least one alternative is stronger. Adding competitors can only maintain or decrease O_q : a positive advantage is conditional on the candidate family, whereas a negative counterexample does not require an exhaustive family. This operational ownership criterion measures relative advantage, not a necessary condition for a concept to participate in computation. Direction specificity requires a positive Effusion effect above random p95 and absolute sham, together with a one-sided 95% lower bound for O_{Effusion} above zero.

Causal-ownership matrix. To test whether another clinical normal better matches the answer pathway, we prospectively cross six Qwen directions (Effusion, Atelectasis, Pneumothorax, Cardiomegaly, Mass, and Nodule) with their six matching questions on 400 new patients. Every question uses the same $\alpha = +0.25$, 119 random directions, and a sham. For question q , ownership requires the diagonal effect $W_{q,q}$ to exceed every other clinical direction, every random direction, and the absolute sham effect. A centered studentized max- T bootstrap gives simultaneous 95% lower bounds over all 30 clinical contrasts. The shared-alias family contains two statistics per direction: mean excess over the matching diagonals on its five off-diagonal questions and signed mean effect on those questions. These give 12 max- T bounds; the effect must also exceed the random alias maximum and maximum absolute sham effect, detailed in Appendix C.

For a family of m statistics, let $\hat{C}_j$ be statistic j 's estimate and C_{bj}^* its value in joint patient-bootstrap draw b , with $B = 5,000$ draws. Write $\tau_j = \text{SD}_b(C_{bj}^*)$ for the bootstrap standard deviation with denominator $B - 1$, held fixed across draws, and $Q_{.95}$ for the empirical 95th percentile with linear interpolation. For $\tau_j > 10^{-12}$, the centered standardized draw Z_{bj} , common critical value $c_{.95}$, and simultaneous lower bound L_j are

$$Z_{bj} = \frac{C_{bj}^* - \hat{C}_j}{\tau_j}, \quad c_{.95} = Q_{.95} \left(\left\{ \max_{1 \leq j \leq m} Z_{bj} \right\}_{b=1}^B \right), \quad L_j = \hat{C}_j - c_{.95} \tau_j. \quad (9)$$

Joint resampling preserves paired conditions; no standard error is re-estimated within a draw. Appendix E specifies the families, degenerate components, and the separate percentile intervals for target-minus-maximum margins.

Answer-encoding crossover and Mass confirmation. A registered crossover on 200 patients of the causal-ownership cohort, with the other 200 as a patient-disjoint calibration set, crosses the original yes/no prompt with A-present and B-present prompts that exchange the finding-to-letter mapping, at $\alpha = \pm 0.25$ with six clinical directions, 20 random directions, and a sham; a question enters only if both coded mappings have clean-answer AUROC lower bounds above 0.5 with at least ten patients per class. An independent confirmation then tests Mass on 200 patients absent from every earlier Qwen cohort, crossing two wordings (*show*, *is*) with explicit yes/no, A-present, and B-present instructions at $\alpha = +0.25$, with 119 random directions, a sham, and 20 clinical comparisons under simultaneous 95% lower bounds. Confirmation requires positive bounds and Mass effects above every random effect and the absolute sham, across all four coded cells or both original-wording cells.

Table 24: **Pilot replication on 600 new patients.** Qwen2.5-VL-7B on NIH ChestX-ray14. Per concept: the concept write $W_{q,q}$, the random 95th percentile (p95), the absolute sham, and its rank among the 120 compared effects; blue shades a negative ownership contrast.

Concept	S	AUROC	$W_{q,q}$	p95	sham	rank	O_q [95% CI]	Competitor	Verdict
Effusion	0.102	0.593	+0.190	0.137	0.017	2/120	-0.071 [-0.080, -0.062]	Nodule	competitor
Atelectasis	0.062	0.566	+0.101	0.297	0.096	55/120	-0.226 [-0.233, -0.219]	Nodule	competitor
Pneumothorax	0.152	0.508	+0.089	0.320	0.113	52/120	-0.102 [-0.108, -0.096]	Effusion	competitor
Cardiomegaly	0.205	0.742	+0.085	0.311	0.158	33/120	-0.219 [-0.227, -0.211]	Atelectasis	competitor
Mass	0.021	0.678	+0.359	0.372	0.092	10/120	-0.169 [-0.178, -0.160]	Effusion	competitor
Nodule	-0.041	0.642	+0.118	0.158	0.055	15/120	-0.072 [-0.081, -0.063]	Effusion	competitor

C Pilot study: results

C.1 A non-generic write effect reproduces

The first finding starts with a successful intervention: Qwen Effusion reproduces beyond random and sham controls before clinical alternatives change its attribution.

Controlled decodability is evaluated at the final visual block each connector consumes: the penultimate CLIP encoder layer for LLaVA and the last of Qwen’s 32 visual blocks. Intervals are 95% patient-cluster bootstrap intervals over 2,000 draws; the control AUROC is the mean over the 20 recurring-type random-label maps, with its 5th–95th percentile spread in parentheses.

Qwen Effusion first satisfies the reader test. Its probe reaches AUROC 0.7742 [0.7488, 0.7993] against a control AUROC of 0.6576 (0.5738–0.7476), a controlled selectivity of 0.1166 (95% CI [0.0897, 0.1428]). LLaVA-1.5-7B reads both of its concepts at its consumed block: Effusion at AUROC 0.7788 [0.7538, 0.8031] with selectivity 0.1139 [0.0875, 0.1404], and Edema at AUROC 0.8009 [0.7648, 0.8329] with selectivity 0.1360 [0.0994, 0.1693], against a control AUROC of 0.6649 (0.5750–0.7450). Original-cohort intervention outcomes maximize the concept-consistent effect over six fixed nonzero doses. Qwen makes a strong write-direction case in the original 200-image cohort: the maximum concept-consistent change is 0.2343 at $\alpha = +0.25$, above the random-direction 95th percentile of 0.0963 and the maximum absolute sham effect of 0.0852.

The response at the locked dose $\alpha = +0.25$ reproduces on the patient-disjoint 400-patient cohort. Effusion changes mean $P(\text{yes})$ by 0.1945, again above the random-direction 95th percentile of 0.0878 and absolute sham effect of 0.0157. None of these patients appears in the original intervention cohort, and the dose was not reselected, supporting a large and reproducible non-generic effect within ChestX-ray14.

C.2 Semantic competition changes attribution, not efficacy

The fixed clinical family provides a harder identification test than non-semantic controls. Nodule changes the Effusion answer by 0.2519 and realizes the point-estimate maximum among the five alternatives. The Effusion ownership contrast O_{Effusion} is -0.0573 . The registered fixed-family endpoint recomputes the maximum inside each of 5,000 patient-bootstrap draws, giving a two-sided 95% CI of $[-0.0694, -0.0450]$ and a one-sided 95% lower bound of -0.0678 . The entire interval is below zero. Descriptively, all five alternative clinical normals move the Effusion endpoint in the same positive direction; Effusion exceeds four by 0.1172–0.1778, while Nodule alone has a larger point estimate. This shared-sign pattern motivates an answer-sensitive-alias hypothesis but cannot distinguish it from question-specific ownership.

Qwen Effusion therefore remains a reproducible direction that changes the answer beyond random and sham perturbations, while its specific semantic attribution fails. The familywise endpoint establishes that at least one fixed clinical alternative is stronger; Nodule is the point-estimate maximizer, not a separately familywise-identified semantic owner.

C.3 Wording changes clinical competition for Mass

In the registered crossover, the four questions that pass capability calibration (Atelectasis, Pneumothorax, Mass, and Nodule) respond more in the component that reverses the raw A-minus-B sign when the finding-to-letter mapping is exchanged, but so do all 20 random directions and the sham, so aggregate mapping response does not separate clinical normals from generic perturbations. Mass is the exception: its clinical margin, the Mass effect minus the largest of the five other clinical effects in finding-present logit units, changes from -0.7181 under the original yes/no prompt to 0.2825 and 0.2900 under A-present and B-present (Table 25, top). The independent confirmation on 200 new patients reproduces the clinical advantage under both *show* coded prompts, with simultaneous lower bounds of 0.1675 and 0.1462 , while both *is* margins are negative and the Mass effects stay below the maximum of the 119 random directions (rank 3 of 120), so neither registered specificity conjunction is met (Table 25, bottom). Mass steering increases Brier error in every cell, and every AUROC-change interval spans zero.

Table 25: **Mass clinical competition across prompts.** Clinical margin per prompt: the Mass effect minus the largest of the five other clinical effects, in finding-present logit units. Top, the discovery crossover; bottom, the registered confirmation on new patients.

Prompt	Clinical margin	95% CI	Mass effect	Random max	Sham
Standard yes/no	-0.7181	$[-0.7956, -0.6412]$	2.8625	3.0412	0.5869
A-present	0.2825	$[0.2269, 0.3406]$	1.6125	1.1650	0.7550
B-present	0.2900	$[0.2300, 0.3506]$	1.7469	1.2581	1.1475
Condition	Clinical margin	Minimum LCB	Mass effect	Random max	
<i>show</i> /A-present	0.2525	0.1675	1.7019	2.2100	
<i>show</i> /B-present	0.2338	0.1462	1.8188	2.2169	
<i>is</i> /A-present	-0.1087	-0.1751	1.9225	2.4750	
<i>is</i> /B-present	-0.2006	-0.2686	2.0019	2.4944	

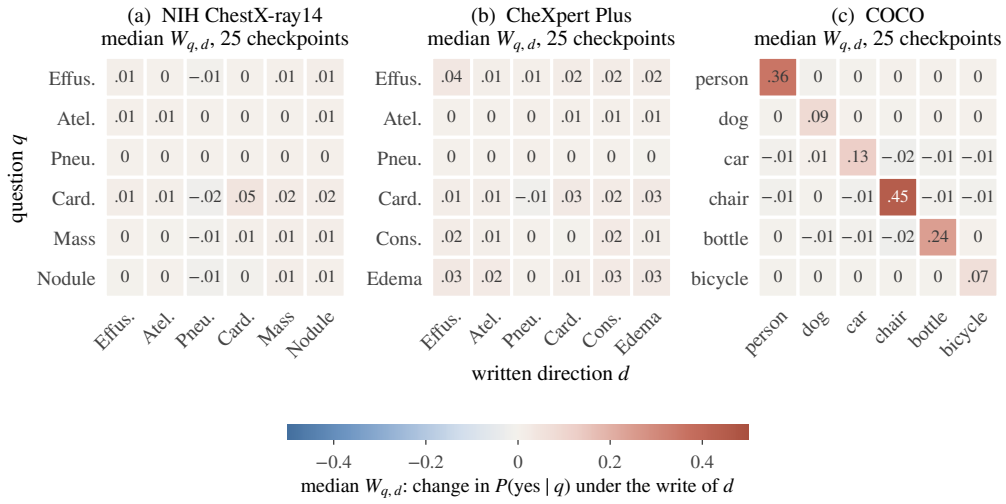


Figure 8: **Write structure across the grid.** Cell-wise median of the write matrix $W_{q,d}$ over completed checkpoints on NIH ChestX-ray14, CheXpert Plus, and COCO. Figure 3 reads the same matrices as flow.

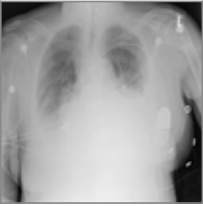
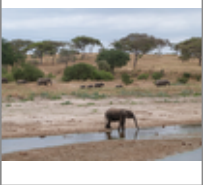
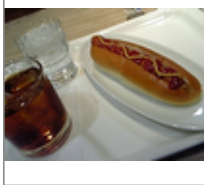
 NIH ChestX-ray14 Qwen2.5-VL-7B	<i>Is there a pleural effusion in this chest radiograph? Answer yes or no.</i> label: Effusion present probe reads 0.51 ✓ clean: No (0.05) Effusion write: No (0.46) ✗ Nodule write: Yes (0.57) competing write moves it more on 9 of 10 such rows	 NIH ChestX-ray14 Qwen2.5-VL-7B	<i>Is there atelectasis in this chest radiograph? Answer yes or no.</i> label: Atelectasis present probe reads 0.52 ✓ clean: No (0.18) Atelectasis write: No (0.30) ✗ Nodule write: Yes (0.55) competing write moves it more on 4 of 4 such rows
 CheXpert Plus Qwen2.5-VL-7B	<i>Is there a pleural effusion in this chest radiograph? Answer yes or no.</i> label: Effusion present probe reads 0.89 ✓ clean: No (0.12) Effusion write: No (0.32) ✗ Consolidation write: No (0.38) competing write moves it more on 71 of 138 such rows	 CheXpert Plus Qwen2.5-VL-7B	<i>Is there consolidation in this chest radiograph? Answer yes or no.</i> label: Consolidation present probe reads 0.77 ✓ clean: No (0.01) Consolidation write: No (0.42) ✗ Effusion write: Yes (0.52) competing write moves it more on 6 of 6 such rows
 CheXpert Plus Lingshu 32B	<i>Is there pulmonary edema in this chest radiograph? Answer yes or no.</i> label: Edema present probe reads 0.89 ✓ clean: No (0.26) Edema write: Yes (0.66) ✓ Effusion write: No (0.49) own write moves it more on 29 of 29 such rows	 NIH ChestX-ray14 Qwen2.5-VL-72B	<i>Is there atelectasis in this chest radiograph? Answer yes or no.</i> label: Atelectasis present probe reads 0.66 ✓ clean: No (0.09) Atelectasis write: Yes (0.73) ✓ Mass write: No (0.15) own write moves it more on 2 of 2 such rows
 COCO Qwen2.5-VL-7B	<i>Is there a dog in this photograph? Answer yes or no.</i> label: no dog probe reads 0.00 ✓ clean: No (0.00) dog write: Yes (0.90) ✓ bicycle write: No (0.04) own write moves it more on 583 of 584 such rows	 COCO Qwen2.5-VL-7B	<i>Is there a chair in this photograph? Answer yes or no.</i> label: no chair probe reads 0.01 ✓ clean: No (0.00) chair write: Yes (1.00) ✓ dog write: No (0.04) own write moves it more on 526 of 526 such rows
 COCO Lingshu 32B	<i>Is there a person in this photograph? Answer yes or no.</i> label: no person probe reads 0.05 ✓ clean: No (0.01) person write: Yes (0.89) ✓ bicycle write: No (0.21) own write moves it more on 265 of 272 such rows	 COCO MedGemma 4B	<i>Is there a car in this photograph? Answer yes or no.</i> label: no car probe reads 0.00 ✓ clean: No (0.00) car write: Yes (1.00) ✓ dog write: No (0.00) own write moves it more on 374 of 500 such rows

Figure 9: **Reading, answering, and writing on single images.** Per image: the question, the report label, the probe’s reading, the clean answer, and the answers under the own write and the strongest competing write at $\alpha = +0.25$. Beside the probe’s reading, a check or cross marks whether it agrees with the label; beside the own write, whether that write moves the answer more than the competitor. The grey line counts, among the block’s test rows the probe reads correctly and the model answers no when clean (on the chest sets, rows labelled present), those with the same ordering.

D Questions, Scores, and Notation

Templates and score. Each concept is asked in six templates that cross two wordings, *is there* (I) and *does the image show* (W), with three answer interfaces, yes/no (Y), A-present (A), and B-present (B). A template is named by its wording and its interface, and the primary template is IY. At the first answer position, ℓ_{yes} and ℓ_{no} are the maximum log probabilities among the valid single-token case and leading-space variants of each answer, and the registered score is $P(\text{yes}) = \sigma(\ell_{\text{yes}} - \ell_{\text{no}})$, where σ is the logistic function. Taking the best valid variant for each answer makes the score insensitive to which capitalisation or leading-space form receives the highest probability. The score measures the relative support for yes versus no at the first answer position, not the accuracy of a generated report. Questions are the rows of W and directions its columns, so changing a direction never changes the question being scored.

Notation. Table 26 collects the notation of the paper.

Table 26: Notation used in the paper.

Symbol	Definition
c, q, d	A concept; the yes/no question about a concept; a write direction. Questions and directions are indexed by the concept they belong to.
cell, block	A checkpoint–concept pair on one dataset; a checkpoint–dataset pair.
$\mathbf{h}_{it}, \mathbf{h}_i$	Output of the consumed block for image i and consumed token t ; its mean over the consumed tokens.
D	Hidden width of the consumed block.
R, s, β_c	Fixed $D \times 512$ Gaussian projection; training-row feature scales; logistic coefficients of concept c ’s probe.
$\hat{\mathbf{w}}_c$	Label direction: the probe’s logit gradient $R \text{diag}(s)^{-1} \beta_c$ at unit length (Equation 2).
a_q	Answer direction: the ridge fit of the clean answer margin $\ell_{\text{yes}} - \ell_{\text{no}}$, lifted in the same way.
α	Signed dose; $ \alpha $ is the write’s norm as a fraction of each token’s norm.
S	Probe selectivity: concept AUROC minus the mean AUROC of 20 random-label control probes (Equation 3).
$W_{q,d}$	Mean change in question q ’s $P(\text{yes})$ when direction d is written.
O_q	Ownership, $W_{q,q} - \max_{d \neq q} W_{q,d}$ (Equation 4).
O_q^m	Ownership computed on changes in the logit margin $\ell_{\text{yes}} - \ell_{\text{no}}$.
κ_d	Column selectivity, $W_{d,d} / \sum_q W_{q,d} $.
$F_{q,d}$	Mean over checkpoints of $\max(W_{q,d}, 0)$, the band width in Figure 3.
$\cos_\Sigma$	Whitened cosine under the training-feature covariance Σ (Appendix A.9).

E Joint bootstrap and simultaneous lower bounds

Equation 9 uses the standard deviation across bootstrap estimates as a fixed scale for each statistic. With $\bar{C}_j^*$ denoting their mean, that scale is

$$\tau_j = \left[\frac{1}{B-1} \sum_{b=1}^B (C_{bj}^* - \bar{C}_j^*)^2 \right]^{1/2}. \quad (10)$$

The standardized deviations are centered on the observed $\hat{C}_j$, not on $\bar{C}_j^*$. For a component with $\tau_j \leq 10^{-12}$, all its Z_{bj} values are set to zero. If every component satisfies that threshold, the critical value is zero. Otherwise, each draw’s maximum includes these zero components, and its empirical 95th percentile uses linear interpolation. All 5,000 draws are retained. The final subtraction $L_j = \hat{C}_j - c_{.95} \tau_j$ still uses the actual τ_j , including for thresholded components.

For the ownership matrix, we resample 400 patients with replacement using seed 20260904. Each draw uses the same patient indices for every question and intervention, preserving all paired baseline differences. The clinical family comprises the 30 fixed contrasts $W_{q,q} - W_{q,d}$ for the six questions and their five alternatives. Each question’s simultaneous ownership lower bound is the minimum of its five contrast bounds. Simultaneous upper bounds use the same critical value with the sign of every contrast reversed. A question has a stronger competitor if any upper bound is below zero. A question is unresolved if it has neither a positive minimum lower bound nor a negative upper bound. The shared-alias calculation uses a separate 12-statistic family: for each of six directions, its mean off-diagonal effect and its mean excess over the five corresponding matched-direction effects. The Mass confirmation applies the same construction to 200 independently selected patients, with seed 20260910 and a separate family of 20 Mass-minus-alternative contrasts across four coded prompt conditions.

Direct target-minus-maximum intervals summarise a different bootstrap quantity. For each draw, we recompute the largest competing mean and subtract it from the target mean. The displayed two-sided percentile intervals use the 2.5th and 97.5th percentiles of these margins. The earlier Effusion specificity supplement uses their 5th percentile as its one-sided lower bound. These intervals are distinct from the simultaneous bounds on the fixed linear contrasts in Equation 9.

The campaign applies the same construction to every block, using 600 test rows and 2,000 shared bootstrap draws of rows (seed 2026090601). The 30 contrasts $W_{q,q} - W_{q,d}$ receive simultaneous 95% bounds from the centred studentised max- T bootstrap. O_q receives a direct percentile interval with the maximum recomputed in every draw. The logit-scale check in Table 12 applies the same rules to margin-scale deltas.

F Reproducibility specification

This appendix specifies each family’s consumed block, the consumed-token mask, the projection and probe, the control constructions, the cohort roles, and the protocol record at the registered protocol’s level of detail. The released code implements each component.

F.1 Consumed loci per family

The consumed block is the last visual block that feeds the consumer. The connector locus is the consumer’s output that enters the language model. We identify the block from each checkpoint’s own configuration when loading the checkpoint. Each block’s preflight record stores the identified block, its hidden dimension, its consumer, and any branch that bypasses it.

- Qwen2.5-VL, Lingshu, and Qwen3-VL: the merger reads the last block of the vision transformer, which has 32 blocks in Qwen2.5-VL and Lingshu, 24 in Qwen3-VL 4B, and 27 in Qwen3-VL 8B and 32B. A Qwen2.5-VL or Lingshu image contributes 576 patch tokens, which the merger joins in 2×2 groups into 144; a Qwen3-VL image contributes at most 1,024. The Qwen3-VL DeepStack mergers, which read earlier blocks, are not written.
- Gemma 3 and MedGemma: the projector reads the last of the 27 SigLIP encoder layers through the tower’s final layer normalisation, and consumes all 4,096 patch tokens of the 896×896 image.
- InternVL3.5: the projector reads the last InternViT encoder layer, of 24 at 8B and 14B and of 45 at 38B; the tower’s final normalisation is an identity in the released configuration. An image contributes 1,024 patch tokens from a single 448×448 tile.
- LLaVA-1.5 and LLaVA-Med: the projector reads the penultimate of the 24 CLIP encoder layers, and the final layer is computed and discarded. An image contributes 576 patch tokens.
- Llama 3.2 Vision: the projector input concatenates the last of the eight global-encoder layers with five intermediate layers of the local encoder, which bypass the consumed block. An image contributes 1,601 tokens, the CLS token and 1,600 patches, from its one real 560×560 tile.

Table 2 lists the consumed-token count of every checkpoint.

F.2 The consumed-token mask

The write in Equation 1 changes only the consumed tokens at the positions the consumer reads at the locus. We derive these positions for each batch from that batch’s processor outputs:

- Qwen2.5-VL, Lingshu, and Qwen3-VL: the tokens of all images in a batch form one sequence in which every image occupies one contiguous span, whose bounds follow from the image’s patch grid. At the connector locus the span is a quarter as long, one merged token per 2×2 patch group.
- Gemma 3 and MedGemma: all 4,096 patch tokens; the tower produces no CLS token. The number of written tokens must equal the number of image placeholder tokens in the language-model input.
- InternVL3.5: the patch tokens of the single 448×448 tile. The CLS token, which the consumer drops, is excluded. The number of written tokens must equal the number of image placeholder tokens.
- LLaVA-1.5 and LLaVA-Med: the 576 patch tokens. The CLS token, which the checkpoint’s feature selection drops, is excluded.
- Llama 3.2 Vision: the 1,601 tokens, CLS and 1,600 patches, of the one real 560×560 tile. The seven padding tokens of each tile and the three padding tiles are excluded, in agreement with the model’s own aspect-ratio and cross-attention masks.

Scoring stops with an error and no fallback when an image span does not exactly cover the locus, a mask length differs from the token count at the locus, an image has no consumed token, or one forward pass reaches the locus more than once.

Preflight check (C) verifies the mask. It captures the locus tensor on a clean pass and a pass steered at $\alpha = +0.25$ along the first concept direction. It reports the maximum absolute difference outside the mask and the maximum absolute changes in the consumer’s output and answer logits. Across all 75 blocks, the change outside the consumed tokens is 0 at both loci. The consumer’s output and answer logits both change. The same records show a bitwise-identical repeated clean pass (check A) and a bitwise-identical $\alpha = 0$ pass (check B) for every block.

F.3 Projection, lift, scaler, and probe

We construct the projection $R \in \mathbb{R}^{D \times 512}$ by drawing standard normal variates once from a PCG64 generator seeded with 0, dividing by $\sqrt{512}$, and storing the result in float32. Its entries are independent $\mathcal{N}(0, 1/512)$ variates. The seed is 0 for every checkpoint, dataset, and locus. D is the family’s hidden dimension. We neither normalise nor orthogonalise the columns of R . We normalise the lifted direction of Equation 2 once to unit ℓ_2 norm in model space. The Gram matrix $R^\top R/D$ has full rank 512 in all 57 blocks that carry the displacement module, with eigenvalues 1.7×10^{-4} – 5.7×10^{-3} and condition number 5.3–33.2.

Projecting the token-mean read vector $\mathbf{h}_i$ gives features $R^\top \mathbf{h}_i$. A featurewise standardiser fitted only on training rows supplies the mean $\boldsymbol{\mu}$ and scale s . Every row is standardised as $(R^\top \mathbf{h} - \boldsymbol{\mu}) \oslash \max(s, 10^{-8})$, where $\oslash$ and $\max$ act elementwise. Each concept probe is an ℓ_2 -regularised logistic regression with inverse regularisation strength $C = 1$. We fit each probe by L-BFGS for at most 2,000 iterations with unweighted classes and seed 0, using training rows with a known label for that concept. The positive-class coefficient fixes the direction’s clinical pole. For each of the two refit seeds, we resample training units with replacement, retaining all rows of each sampled unit at its multiplicity. We then refit the scaler and six probes using the same projection. The random family, shams, and control probes remain at seed 0.

F.4 Control constructions

Random directions. We draw one $119 \times D$ standard-normal matrix using a PCG64 generator seeded with 0 and normalise each row to unit ℓ_2 norm. Every question in the block uses the same 119 directions.

Coordinate-permutation sham. Immediately after the random draw, the same generator produces one permutation of $\{0, \dots, D-1\}$ per concept, in protocol concept order. We form concept c 's sham by rearranging the coordinates of $\hat{\mathbf{w}}_c$ with concept c 's own permutation. This preserves the direction's coordinate multiset and unit norm. We write each question with its own sham.

Matched random-label probes. Each row has a type: view $\times$ sex $\times$ age decade on the chest sets, and aspect-ratio $\times$ area buckets on COCO. For control seed $k \in \{0, \dots, 19\}$, we traverse the types in sorted order and use a PCG64 generator seeded with k to draw one label per type uniformly from $\{0, 1\}$. If all types receive the same label, we flip the first. Every row of a type receives that type's label. We give each control probe the concept probe's capacity and nuisance structure by fitting it on the same training rows and known-label subset with the same hyperparameters. A block fits control probes when its training rows span at least eight types. Selectivity is Equation 3. Its bootstrap over calibration units retains a draw when the real AUROC is finite and at least ten of the twenty control AUROCs are finite. We grade a cell for readability when at least 1,900 of the 2,000 draws are retained.

F.5 Cohort roles

Table 27 lists each role's size, its independent units, and its use. Every chest image is a frontal study. One role holds more than one row per unit: the NIH training role, which takes every study of each training patient, 18,212 studies from 5,783 patients, 2,655 of whom contribute more than one and one of whom contributes 125. Every other role of every dataset holds one row per unit, and no unit appears in two roles of a dataset. We use the calibration role only to determine readability and answerability. We neither fit probes nor score writes on it. Every write uses the test role.

Table 27: **Cohort roles.** Rows per role, with independent units in parentheses (patients on the chest sets, images on COCO), and the role's use. The last two rows give the largest rows per unit and the largest overlap between two roles.

Role	NIH ChestX-ray14	CheXpert Plus	COCO	Use
train	18,212 (5,783)	20,000 (20,000)	20,000 (20,000)	Scaler, six concept probes, twenty control probes, the two nuisance probes of the chest sets, and the answer-direction ridge. No row of this role is scored.
preflight	16 (16)	16 (16)	16 (16)	The seven acceptance gates, at both loci.
calibration	400 (400)	400 (400)	400 (400)	Clean pass at $\alpha = 0$: probe readability and answerability.
test	600 (600)	600 (600)	600 (600)	Every write matrix, dose, refit, template, locus, and addendum module.
valid	–	200 (200)	–	The regrade on radiologist consensus labels (Table 20).
rows per unit	125 (2,655 units)	1 (0)	1 (0)	Largest number of rows one unit contributes, with the units contributing more than one; the NIH training role is the campaign's only role above one.
shared units	0	0	0	Largest number of units two roles of the dataset share, over all 22 role pairs of the three datasets.

F.6 Protocol record and amendments

Each block writes a run record containing the protocol identifier `cf-transfer-v1`, a run identifier naming the checkpoint and dataset, the evaluation code version that produced the block, the pinned checkpoint and processor revisions, the library versions, the numeric precisions, and the resolved processing settings, including the rendered chat template. All 75 blocks carry the protocol identifier `cf-transfer-v1`. The protocol file registers the code version it was frozen against.

Table 28 lists every addition to the protocol file after the first wave, and the date the file records for it. Every analysis the paper reports that is not one of the planned modules appears here, with the

date the file gives it. The seven planned modules have no amendment date. Each amendment entry names its own rows, references, and decision rule before it names a result, and the robustness tables in Appendix A.8 identify the blocks that carry each one. A1 and A2 also carry the identifiers of the execution log.

Table 28: **Protocol amendments.** Every addition to the registered protocol file after the first wave, with the date the file records for it. Each date precedes the first outcome shard of the module it registers.

Amendment	Date	What it adds
A1	2026-09-14	Per-block primary template: the single-template modules score the first eligible template in the order (IY, IB), so a checkpoint whose yes/no template fails the image-free mapping gate is graded on its B-positive template instead of losing its grade. Llama 3.2 Vision 11B re-ran on all three datasets.
A2	2026-09-14	CheXpert gains the dose sweep, the two probe refits, the connector locus, the connector-locus calibration, and the five additional templates, with Effusion and Edema as its two template concepts, so the second chest set carries the same modules as the first.
A3	2026-09-14	Four alternative direction constructions per concept: difference of class means, Haufe pattern, orthogonalised normal, and residualised normal.
A4	2026-09-14	Extra clinical directions for every additional dataset label with at least 100 known positives and negatives, added to the competitor family.
A5	2026-09-14	Softmax-weighted and top-quarter token weights for the six logistic directions.
A6	2026-09-14	The write grid rescored with fp32 weights and forward pass, and in bf16 at batch size one.
A7	2026-09-14	The answer direction: a ridge regression of the clean answer margin on the same projected, train-scaled features.
A8	2026-09-15	The difference-of-means and pattern vectors lifted as displacements through the minimum-norm preimage.
A9	2026-09-15	Three non-clinical attribute directions written with the six clinical directions as one nine-direction family.
A10	2026-09-15	The answer directions fitted on the primary template written under the five held-out templates.
A11	2026-09-15	The write grid regraded on the CheXpert validation split with radiologist consensus labels.
A12	2026-09-16	The crossed tower and reader: one checkpoint’s vision tower loaded behind the other’s connector and language model, and all four combinations of tower and reader graded on the same rows.
A13	2026-09-16	The identical-tensor replay: the consumed block’s stored token tensors fed to every reader of a shared-tower group.
A14	2026-09-16	The four semantic endpoints: the negated question, the counterbalanced two-finding forced choice, and the finding word’s log-probability in a report continuation, each with its own clean baseline, sham, and random family, together with the incremental-validity test of the ownership contrast.
A15	2026-09-16	The fine-grained object family: six further COCO categories chosen by the rule stated below, fitted, written, and graded as their own six-direction family in the same blocks and rows.
A16	2026-09-16	The clean pass of the fine-grained categories on the calibration rows, so their clean-answer AUROC is measured on the same rows and by the same rule as every other cell and the difficulty matching compares like with like.
A17	2026-09-16	The matched attribute reference: the protocol’s own 119-direction random family written against the three attribute questions. Gives attribute and clinical questions the same steering reference.
A18	2026-09-16	Three phrasings of each attribute question scored clean on the calibration rows, with the selection rule and its reason recorded per cell.
A19	2026-09-16	The expert-label refits: the six concept probes refitted on the radiologist-labelled films, cross-fitted over patients, and written on the test rows.
A20	2026-09-16	Projection seeds 1 and 2: the probes, the random family, and the sham redrawn in two further random projections and regraded against that projection’s own references.

The fine-grained category selection rule. The rule that picks the six fine-grained COCO categories of amendment A15 is stated in the protocol file in full, and it reads on the labels alone. A

category is eligible when its training positives are at least as many as the weakest of the six existing concepts, when it has at least ten positives among the 400 calibration rows and at least ten among the 600 test rows, and when its mean relative instance area is below the median of the six existing concepts. Eligible categories are ranked by calibration positives in descending order, ties by mean relative instance area in ascending order, and remaining ties by category identifier; the first six are taken. The rule uses the label counts and the instance areas of the fixed cohorts and nothing else: no probe, no answer, and no write enters it. The protocol file carries the rule under the date of amendment A15, and the module's first outcome shard is later than that date, so the six categories and the two matching windows were fixed before any of them was scored.